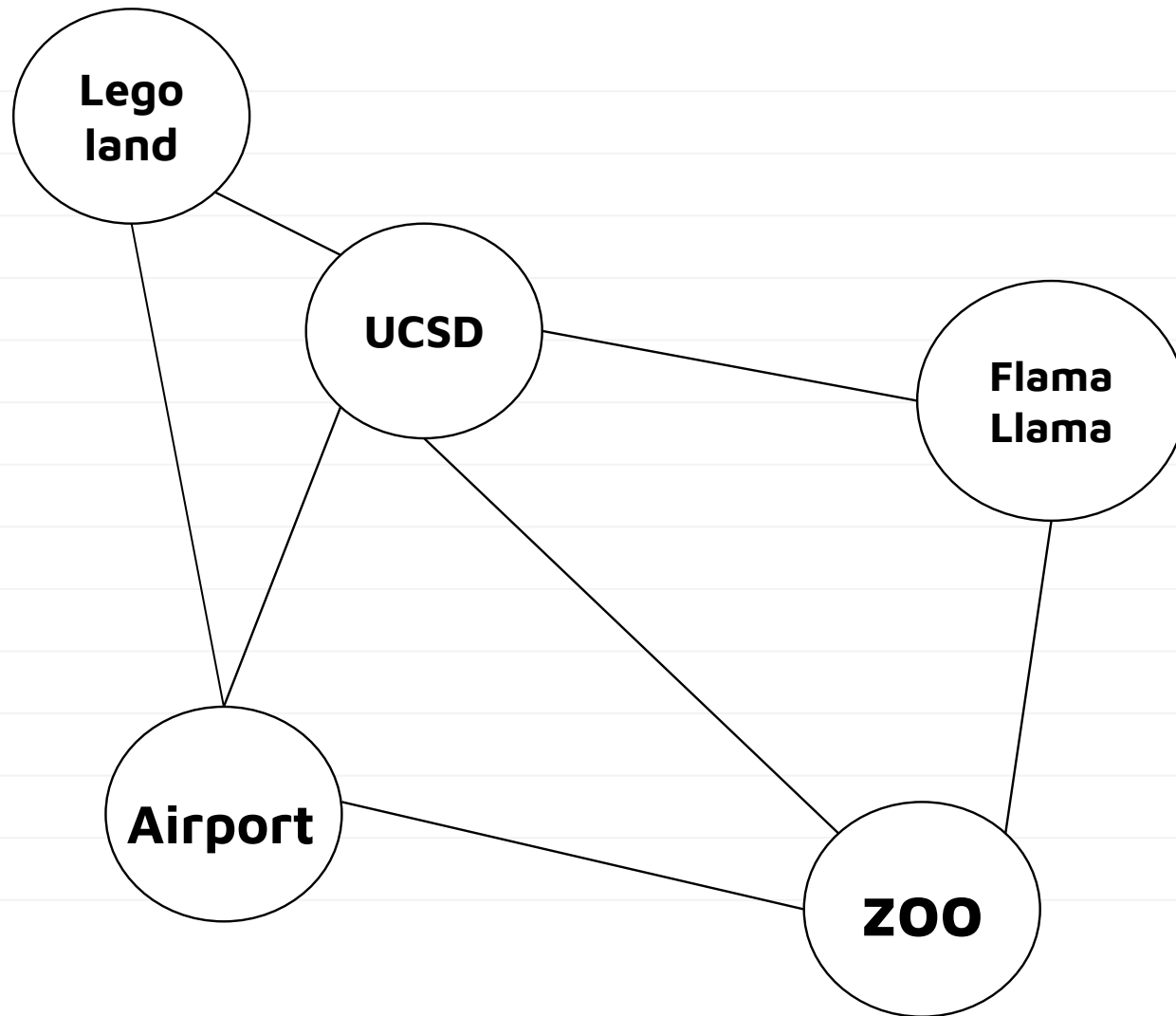


DSC 40B

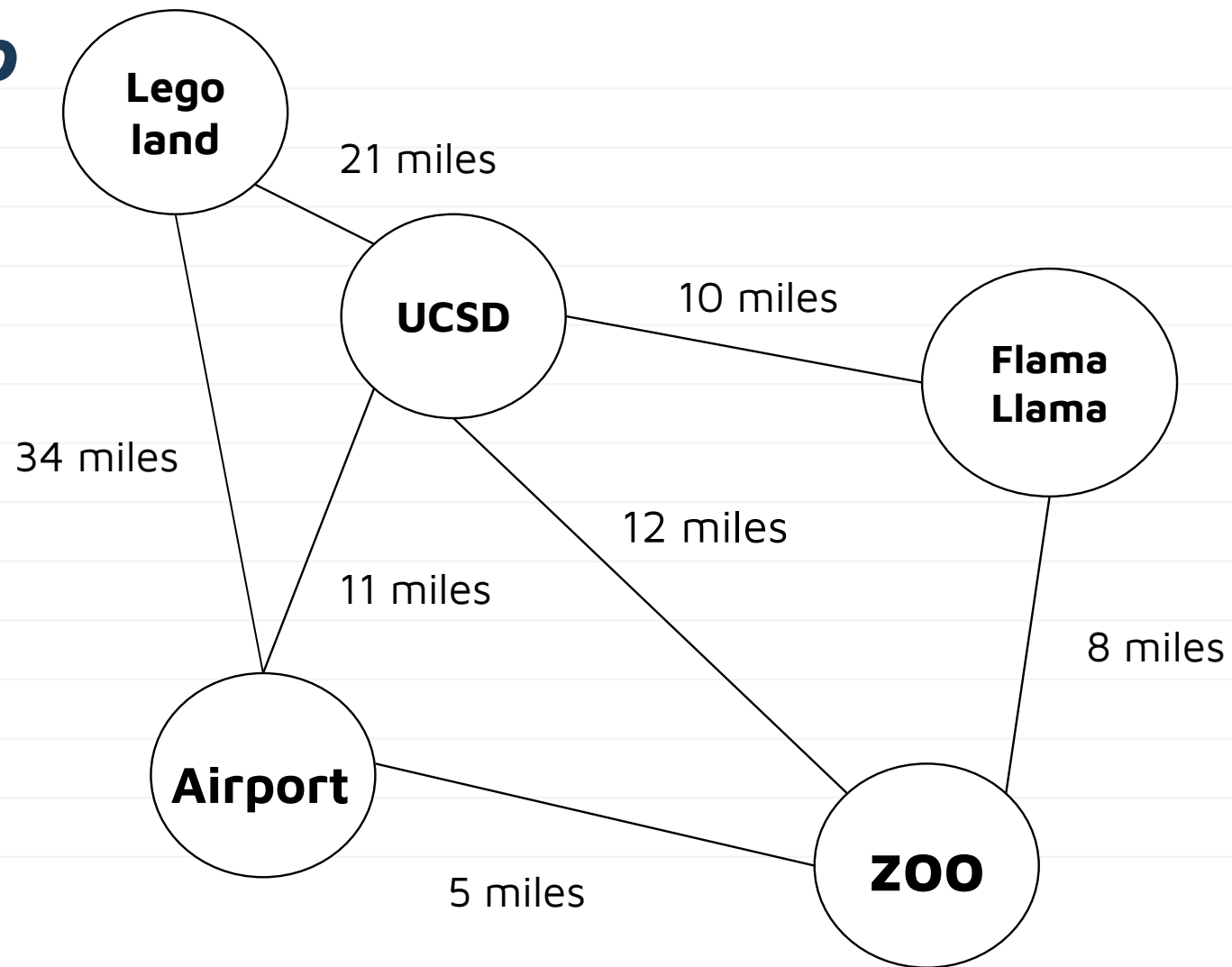
***Lecture 22. : Shortest
Paths in Weighted
Graphs. Dijkstra***

Shortest Paths in Weighted Graphs

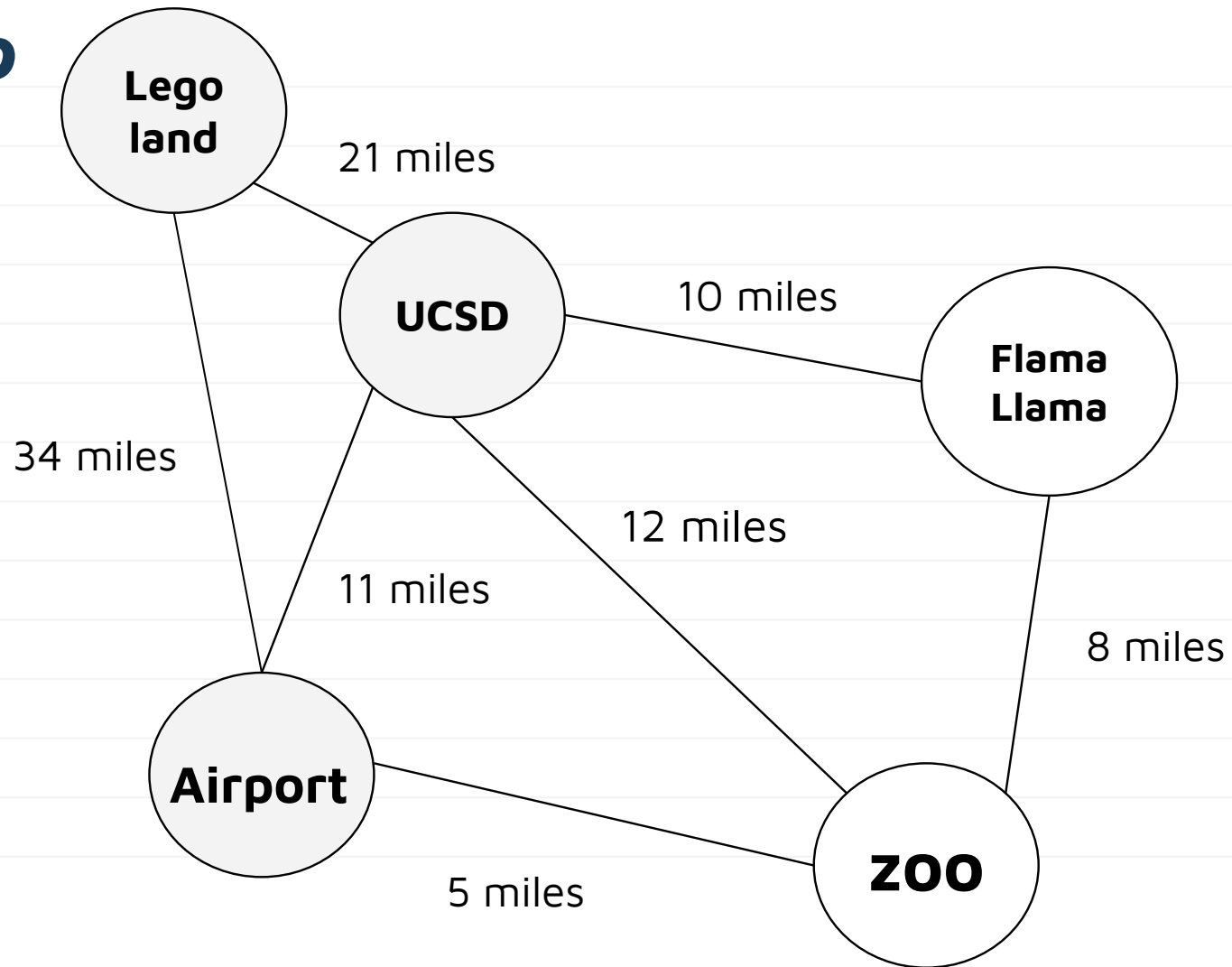
Map



Map



Map



Weighted Graphs

- An **edge weighted** graph $G = (V, E, \omega)$ is a triple where (V, E) is a graph and $\omega : E \rightarrow \mathbb{R}$ maps each edge to a **weight**.
- Can be directed or undirected.
- In general, weights can be positive, negative, zero.
- Many uses, such as representing **metric spaces**.

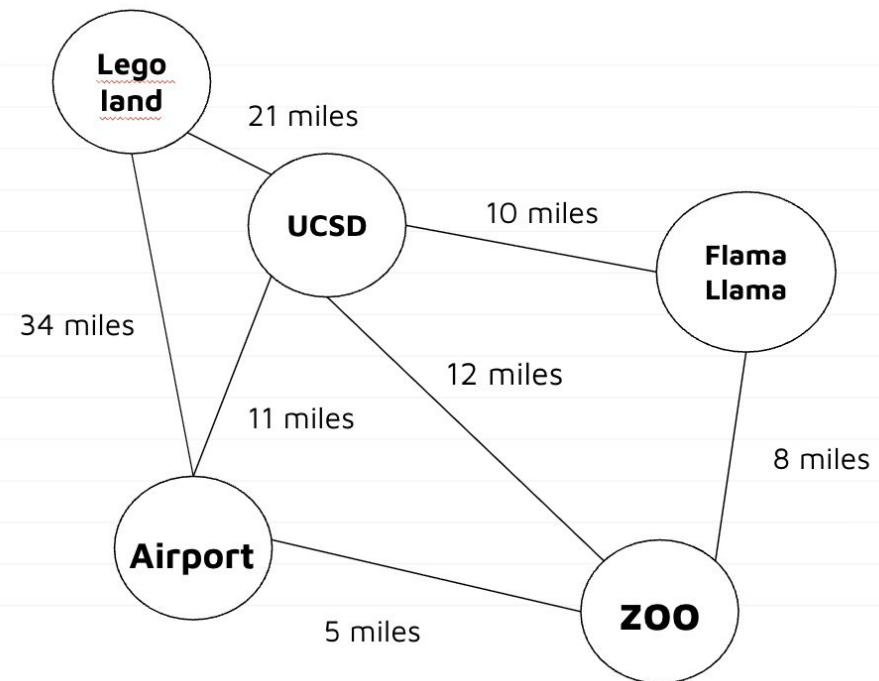
Path Lengths

- The **length** of a path in a **weighted** graph (usually) refers to the **total weight of all edges** in the path.

Example:

(LegoLand, Airport, Zoo)

$$34 + 5 = 39$$

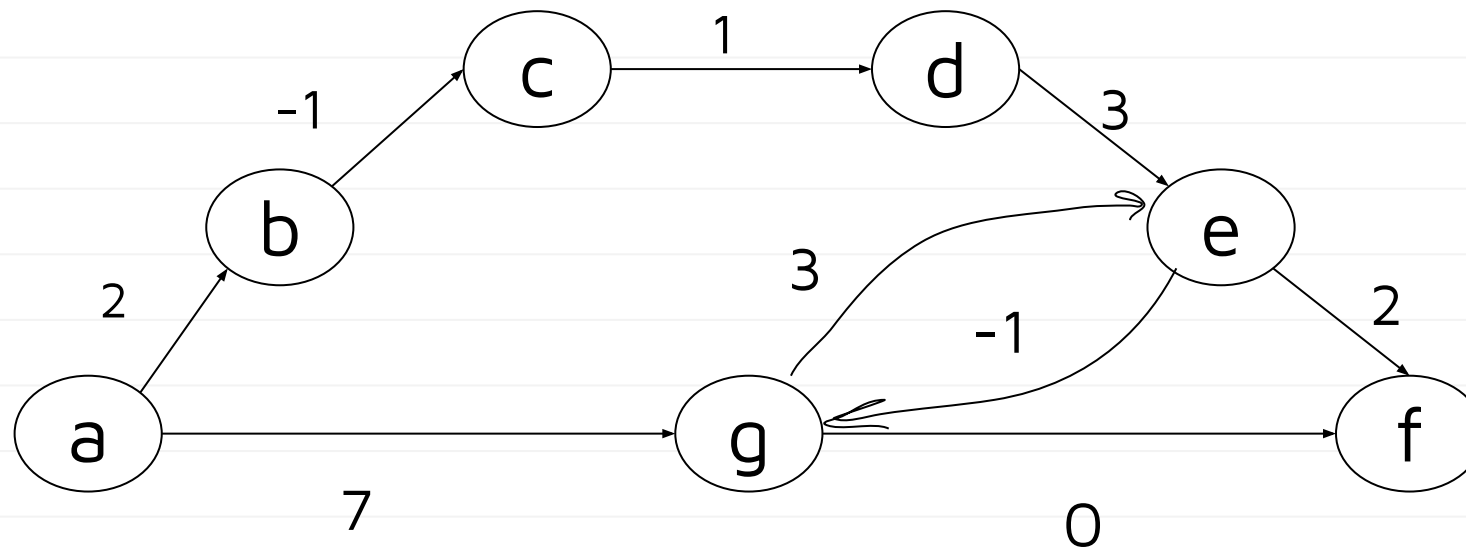


Shortest Paths

- A **shortest path** between u and v is a path between u and v with **minimum** length.
- In other words, **minimum total weight**.

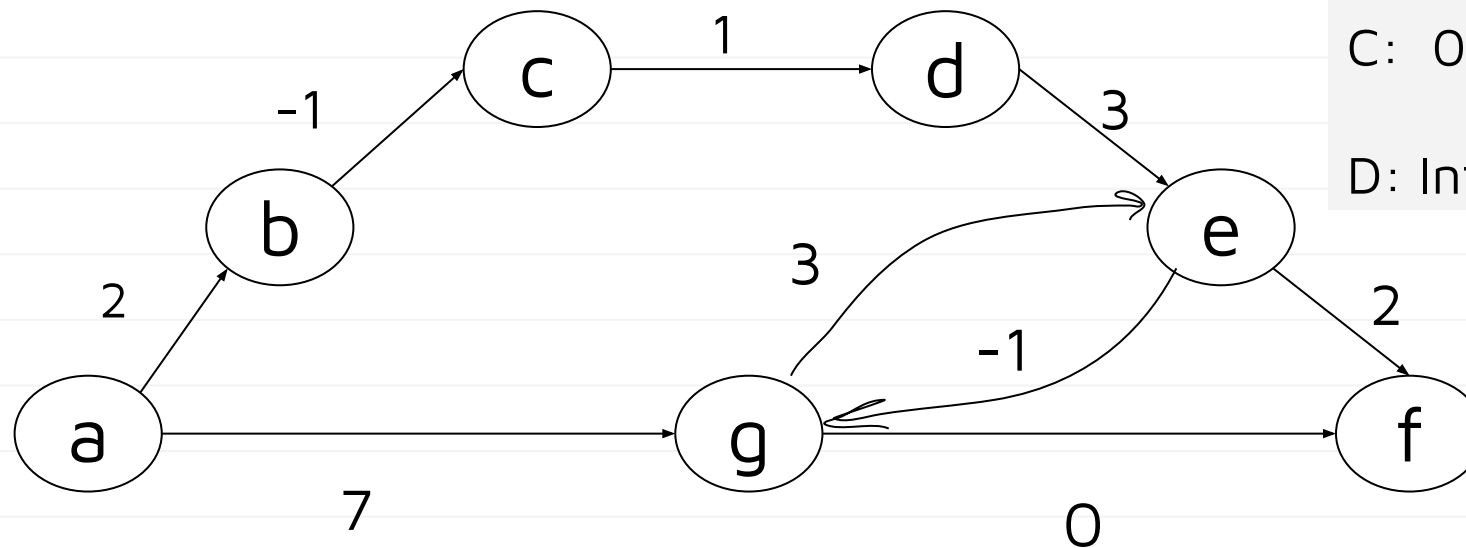
Example

What is the **shortest** path from **a** to **f**?



Example

What is the **shortest** path from **a** to **f**?



A: Some positive number

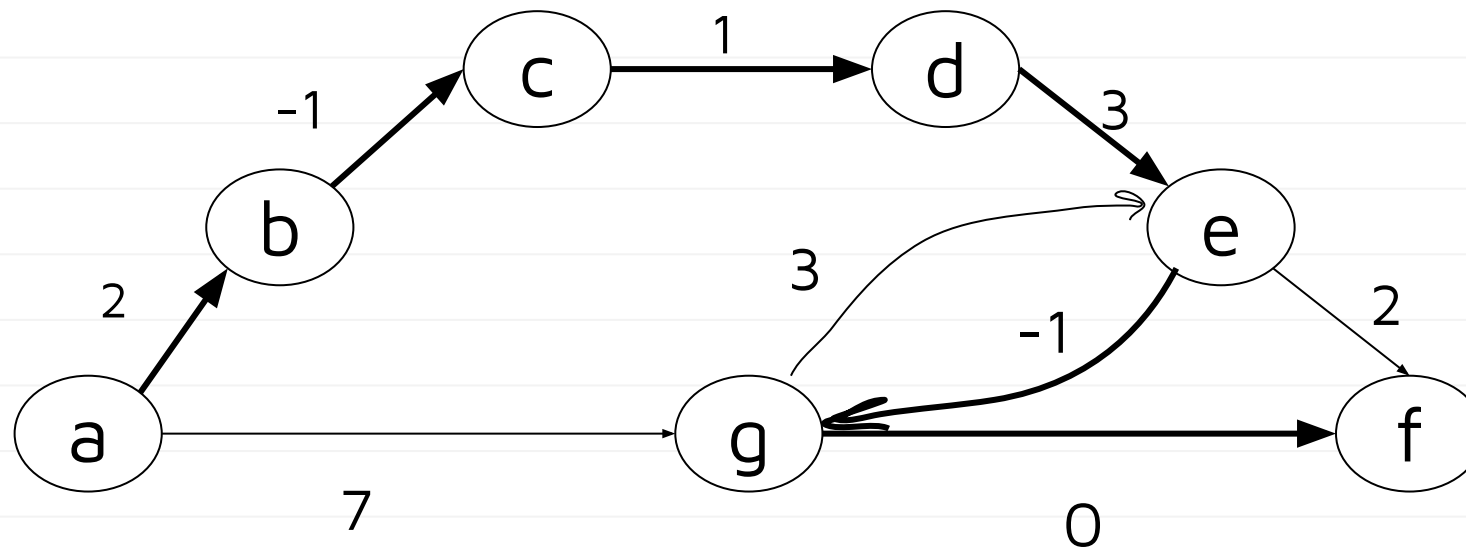
B: Some negative number

C: 0

D: Inf. loop

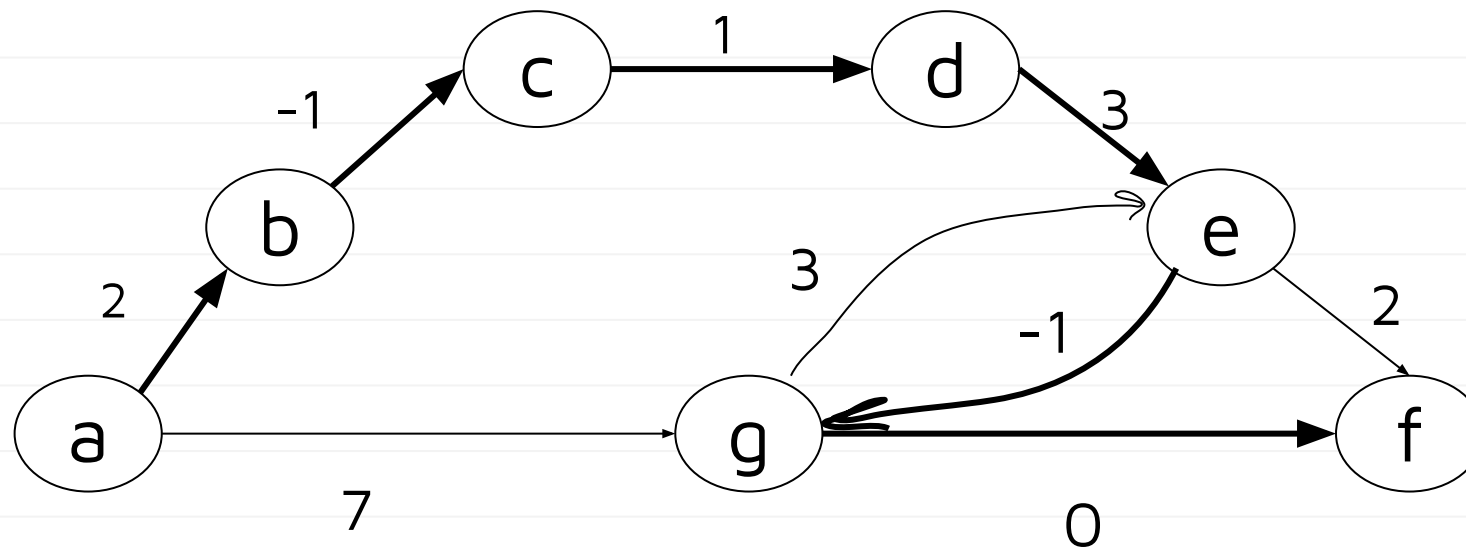
Example

What is the shortest path from **a** to **f**?



Example

What is the shortest path from **a** to **f**?



Path: a, b, c, d, e, g, f

Length: $2 - 1 + 1 + 3 - 1 + 0 = 4$

Today (and next time)

- How do we find shortest paths in weighted graphs?

Idea #0

- Does BFS work?
 - **No**, not really. *Only if all weights are the same.*
- Can we “**convert**” a weighted graph to an unweighted one?

Idea #0. How can we convert?



Idea #0. When will it fail?



Idea #0

- **Step 1:** “Convert” weighted graph to unweighted one with dummy nodes.
- **Step 2:** Call BFS on this new graph.

Idea #0

- Very **inefficient** for large weights.
- What if edge weights are floats, or negative?

Ideas #1 and #2

- We'll look at two algorithms: **Bellman-Ford** and **Dijkstra's**.
 - **Input**: weighted graph, source vertex s .
 - **Output**: shortest paths from s to *every other* node.
- Both work by:
 - keeping track of shortest known path (**estimates**).
 - iteratively **updating** these until they're correct.

Shortest Path Estimates

- Bellman-Ford and Dijkstra's keep track of the shortest paths found so far; we call these the **estimated shortest paths**.
- For each node u , remember u 's:
 - predecessor in estimated shortest path;
 - distance from source s in estimated shortest path.
- **Key:** estimated distance will always be $\geq$ actual distance.

Updates : operations on an edge

- Both algorithms work by iteratively **updating** their estimates.
- On each iteration, consider a **new** edge (u, v) .

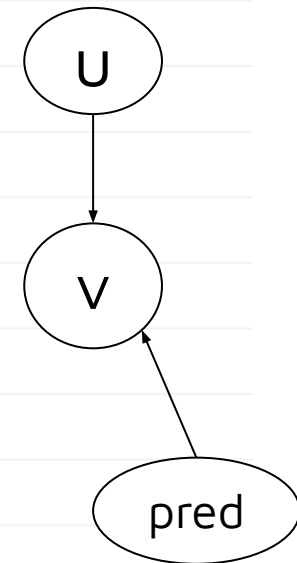
Ask: is the **best** known shortest path from

source $\rightarrow \dots \rightarrow u \rightarrow v$

shorter than the best known shortest path from

source $\rightarrow \dots \rightarrow \text{predecessor}[v] \rightarrow v$?

- If it is, we have discovered a shorter path to v .



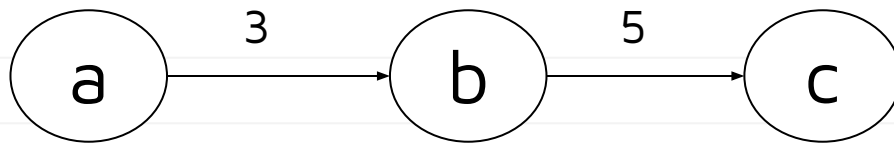
Dijkstra's Algorithm

Dijkstra's Algorithm

All weights are positive.

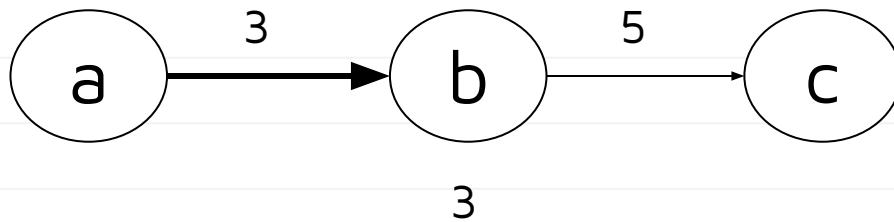
Dijkstra's Algorithm

- Intuition



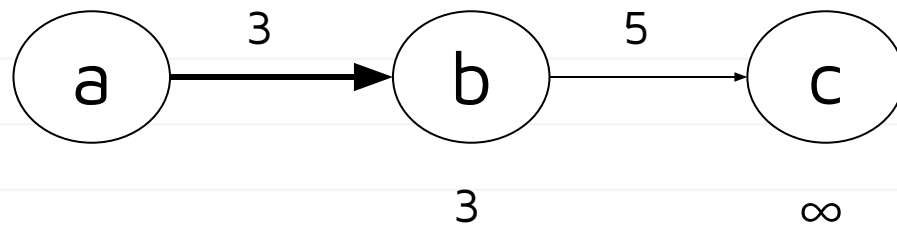
Dijkstra's Algorithm

- Intuition



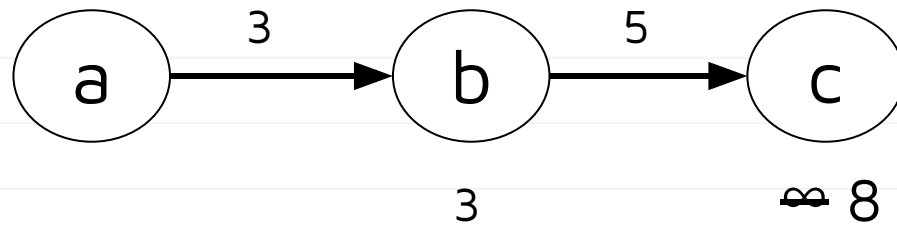
Dijkstra's Algorithm

- Intuition



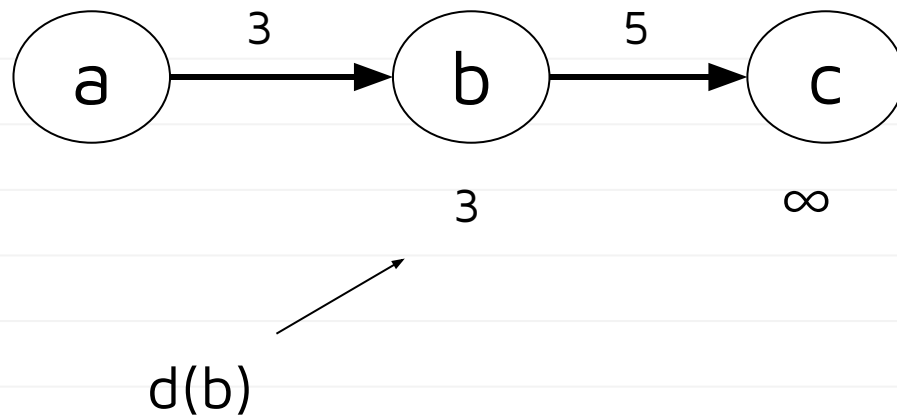
Dijkstra's Algorithm

- Intuition



Dijkstra's Algorithm

- Intuition

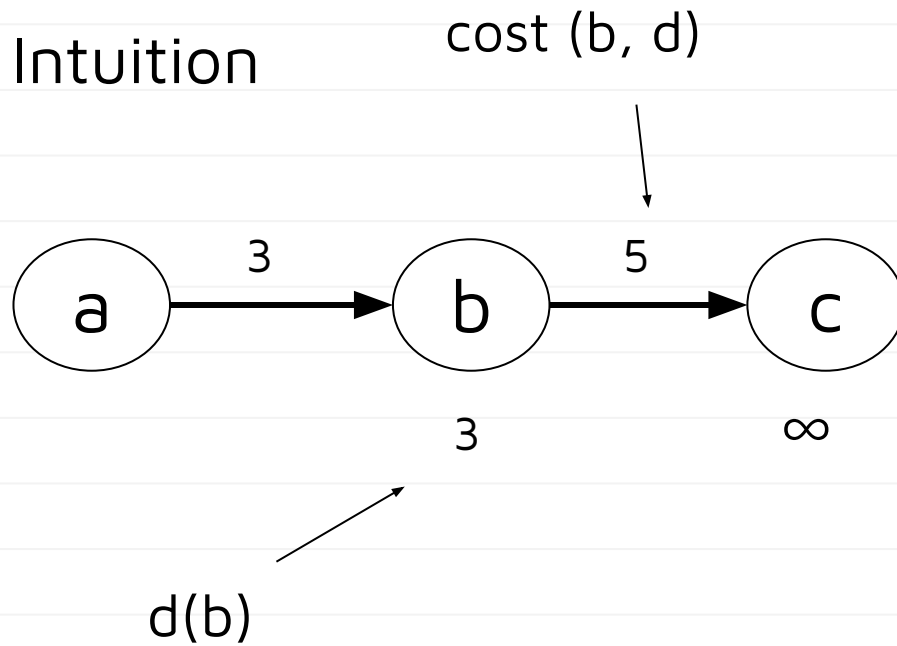


Update Rule:

if $d(b)$

Dijkstra's Algorithm

- Intuition

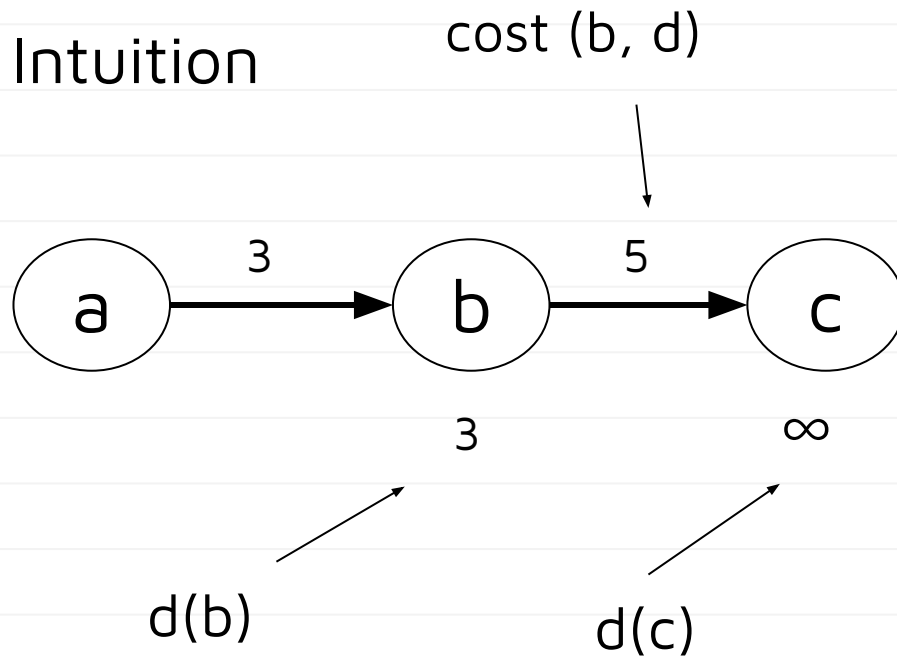


Update Rule:

if $d(b) + \text{cost}(b, c) < d(c)$

Dijkstra's Algorithm

- Intuition

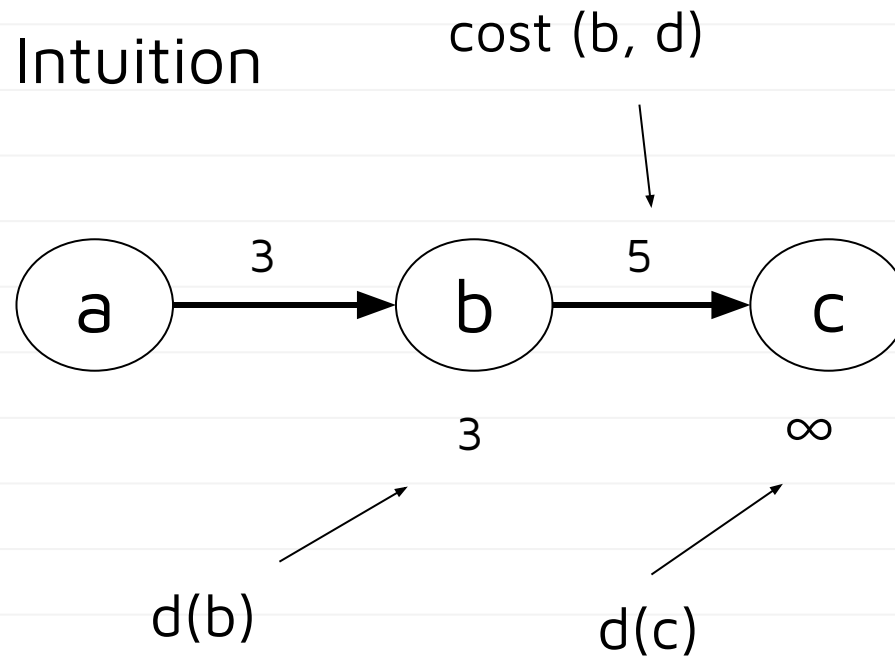


Update Rule:

if $d(b) + \text{cost}(b, c) < d(c)$:

Dijkstra's Algorithm

- Intuition

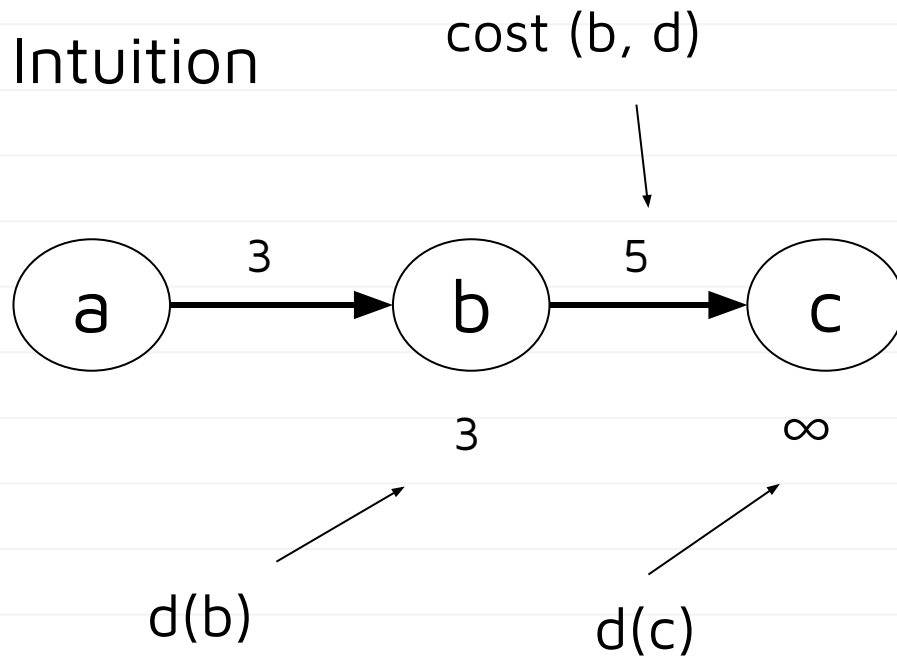


Update Rule:

if $d(b) + \text{cost}(b, c) < d(c)$:
 $d(c) = d(b) + \text{cost}(b, c)$

Dijkstra's Algorithm

- Intuition



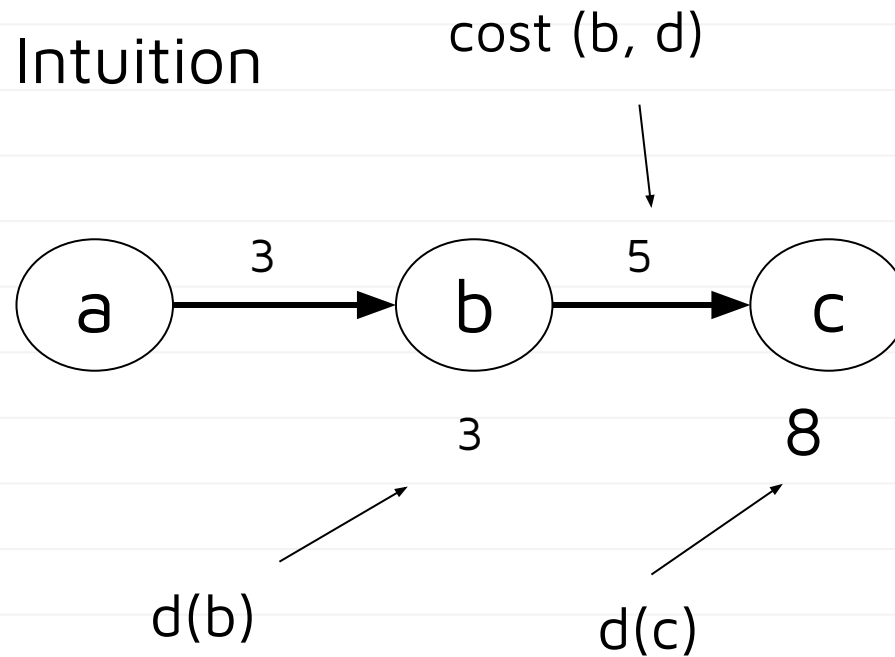
Update Rule:

if $d(b) + \text{cost}(b, c) < d(c)$:
 $d(c) = d(b) + \text{cost}(b, c)$

If $3 + 5 < \infty$:
 $d(c) = 8$

Dijkstra's Algorithm

- Intuition

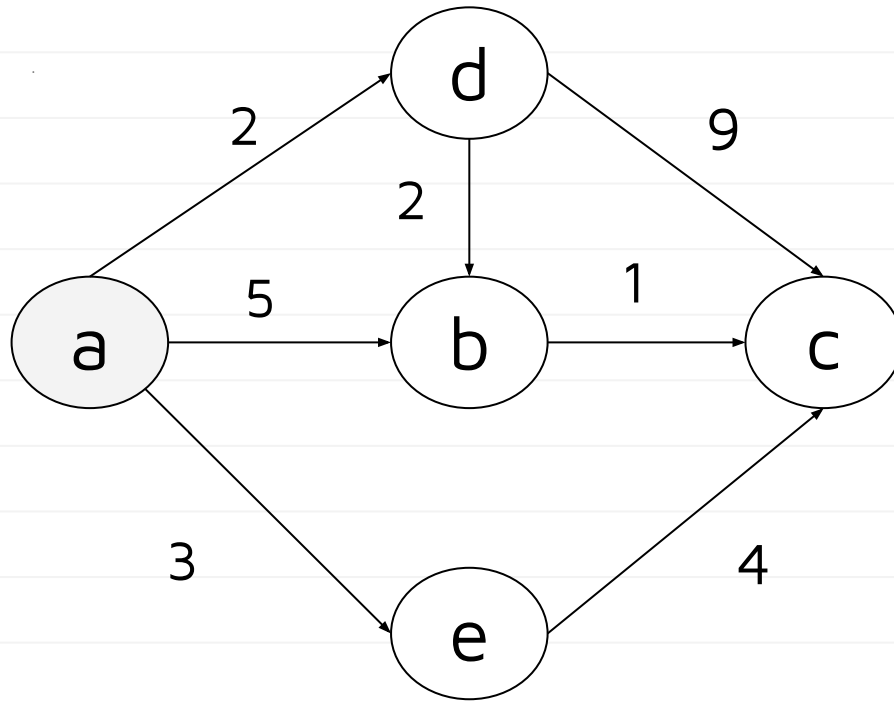


Update Rule:

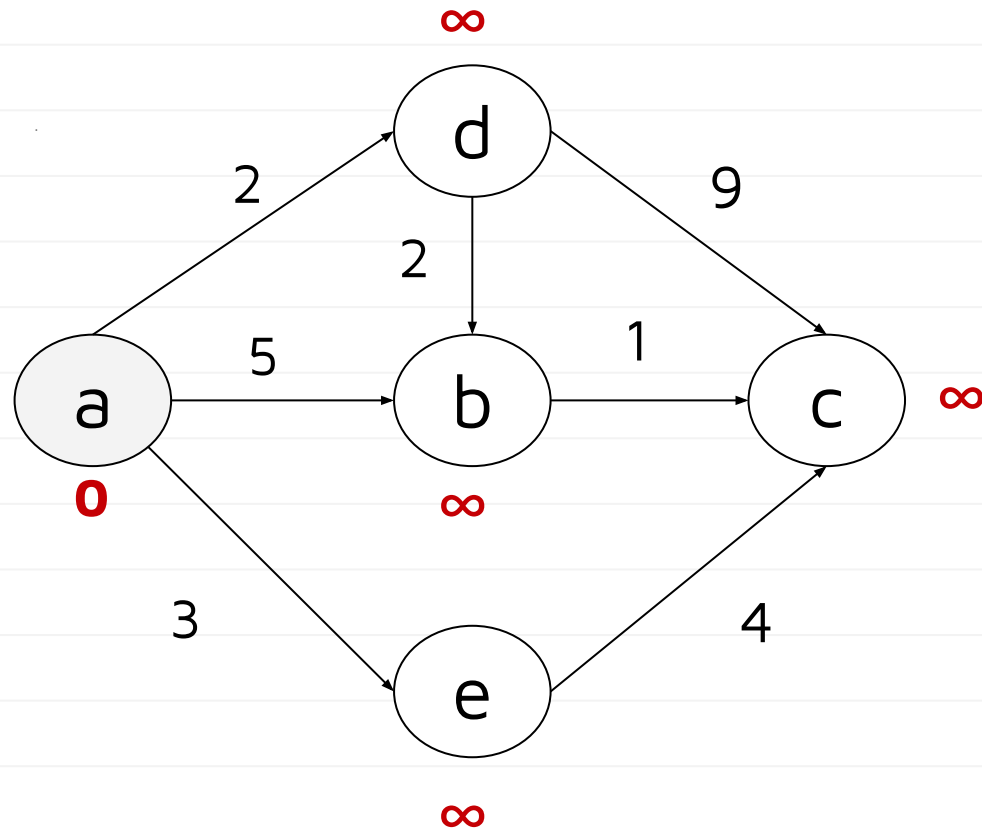
if $d(b) + \text{cost}(b, c) < d(c)$:
 $d(c) = d(b) + \text{cost}(b, d)$

If $3 + 5 < \infty$:
 $d(c) = 8$

Dijkstra's Algorithm



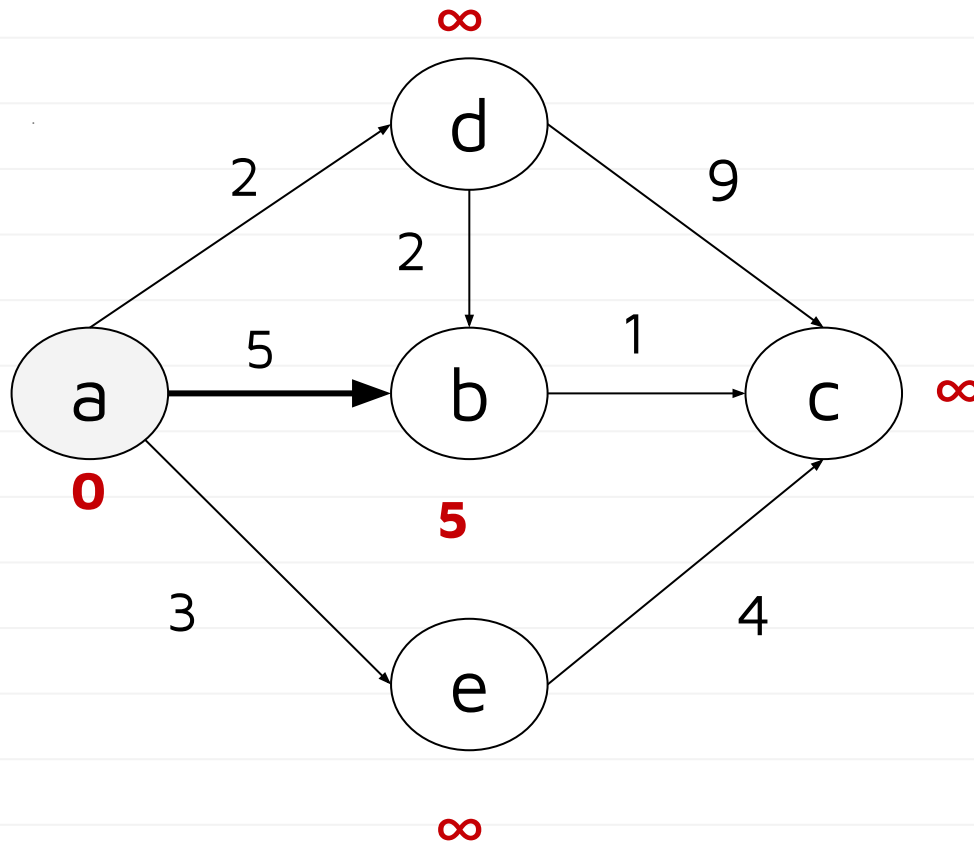
Dijkstra's Algorithm



Dijkstra's Algorithm

Update rule:

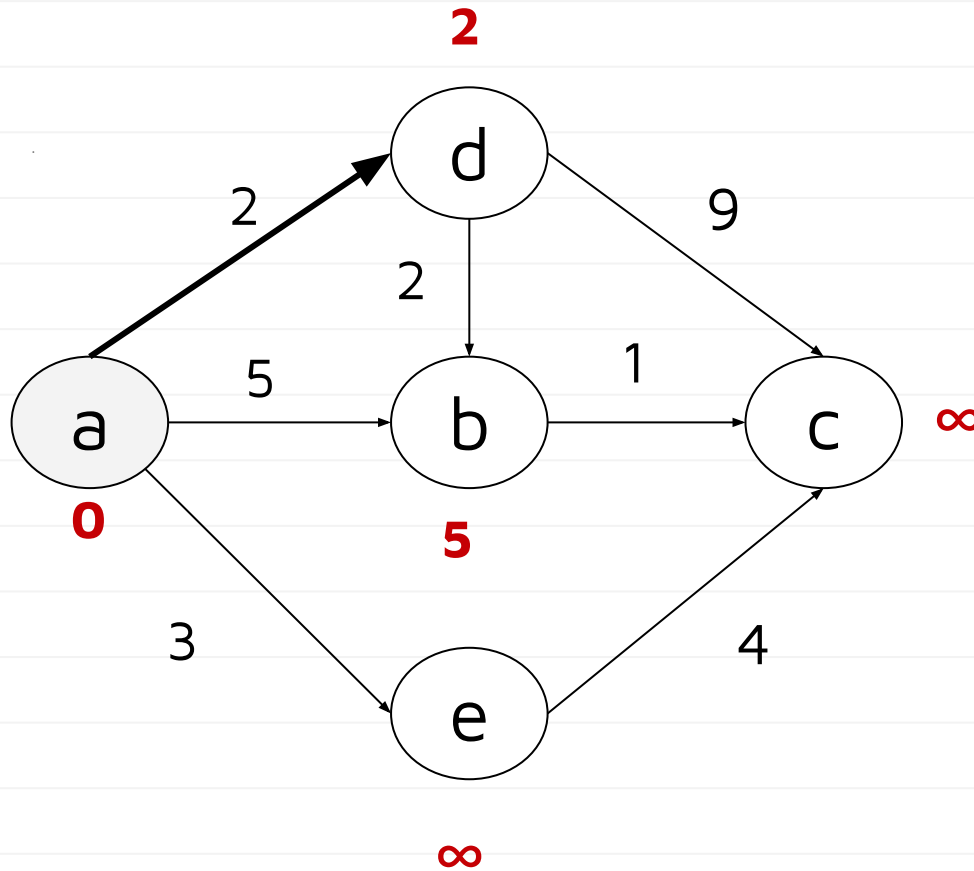
If $0 + 5 < \infty$, update



Dijkstra's Algorithm

Update rule:

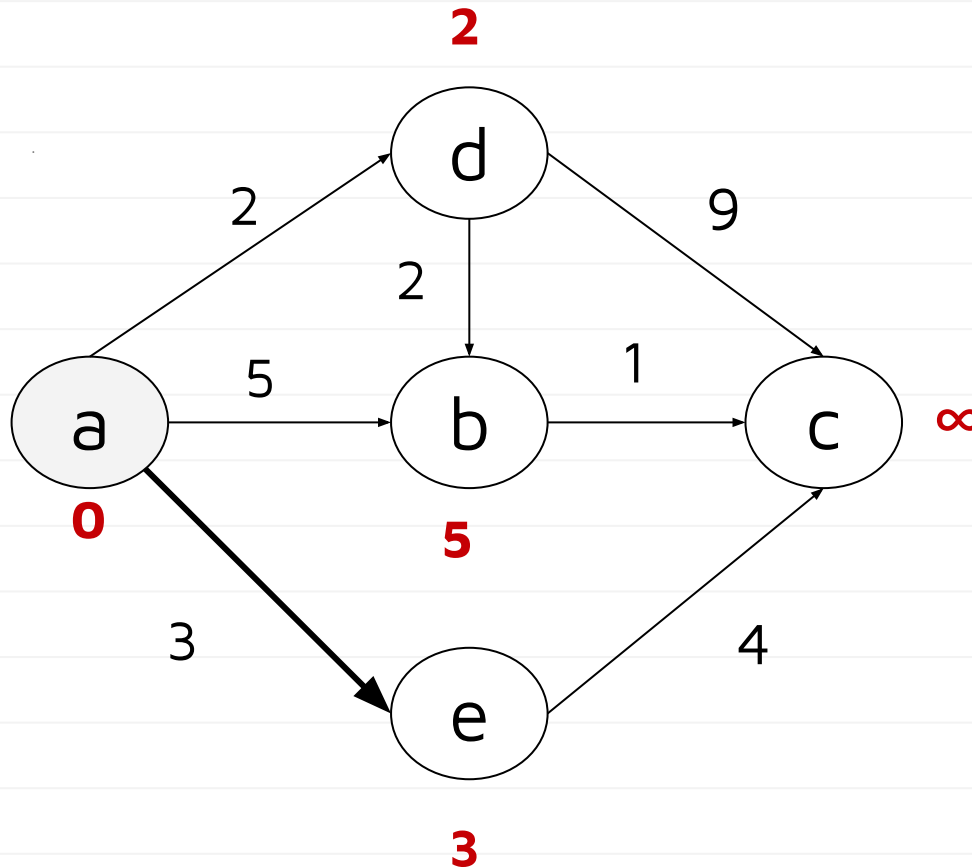
If $0 + 2 < \infty$, update



Dijkstra's Algorithm

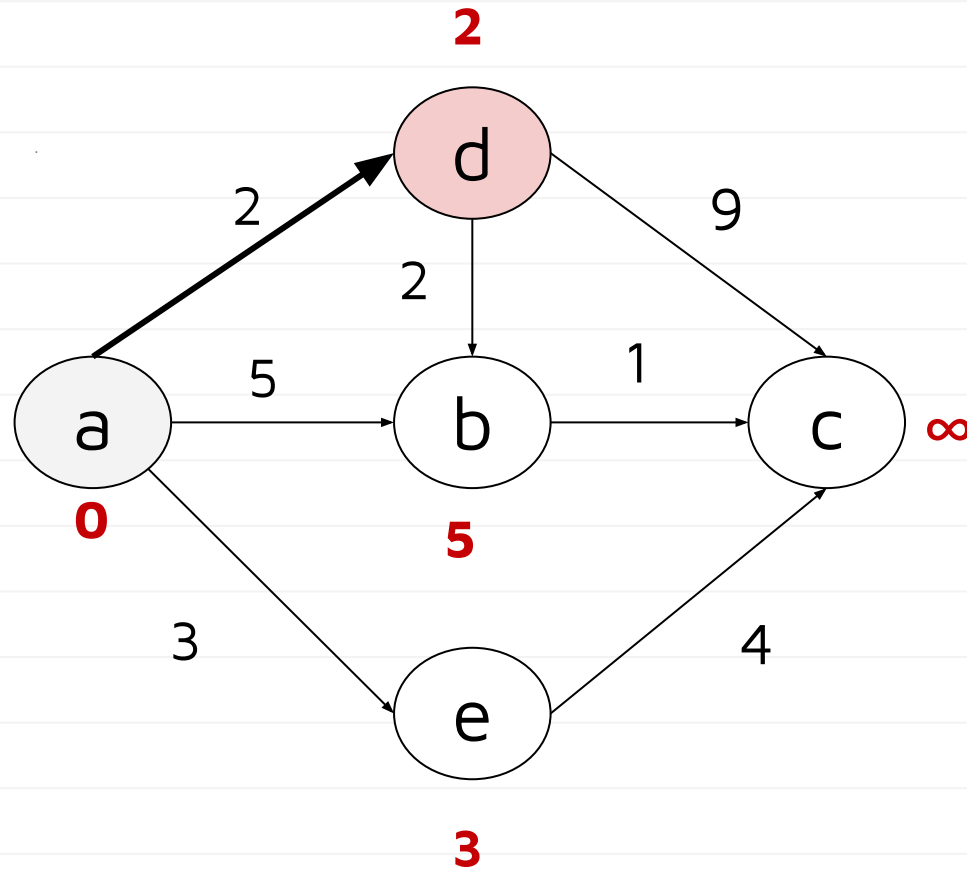
Update rule:

If $0 + 3 < \infty$, update

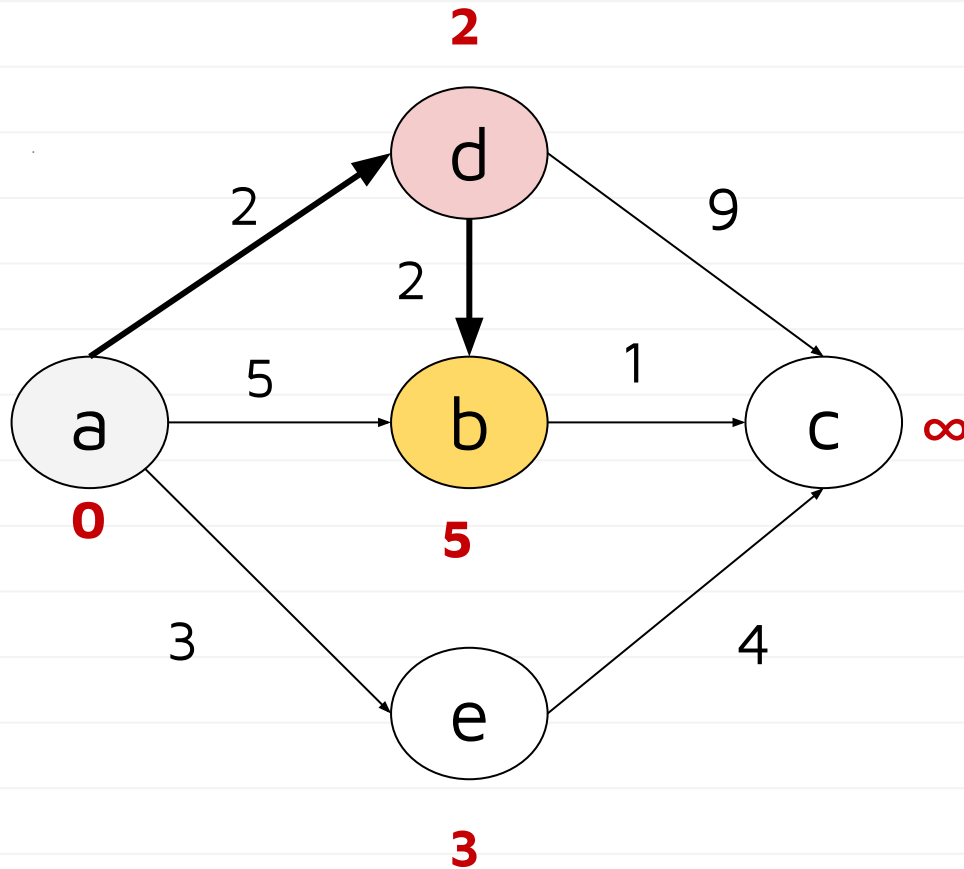


Dijkstra's Algorithm

Select the **shortest** path: 2



Dijkstra's Algorithm



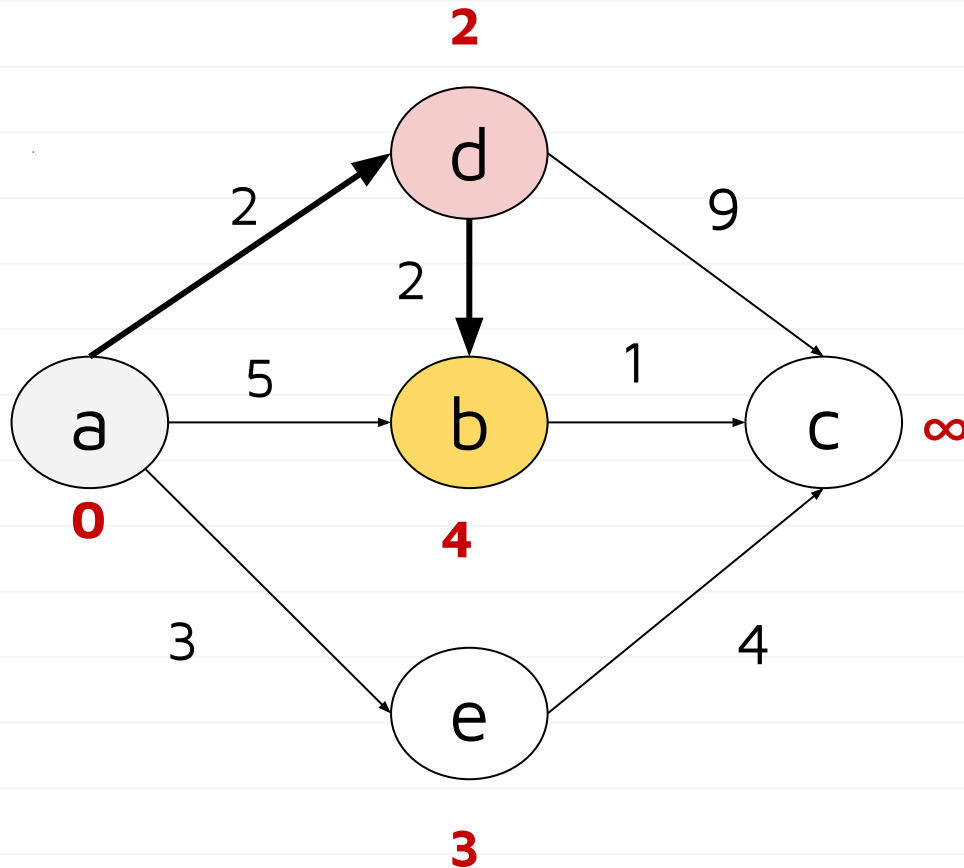
Select the **shortest** path: 2

Update rule:

If $2 + 2 < 5$, update

if $d(u) + \text{cost}(u, v) < d(v)$:
 $d(v) = d(u) + \text{cost}(u, v)$

Dijkstra's Algorithm



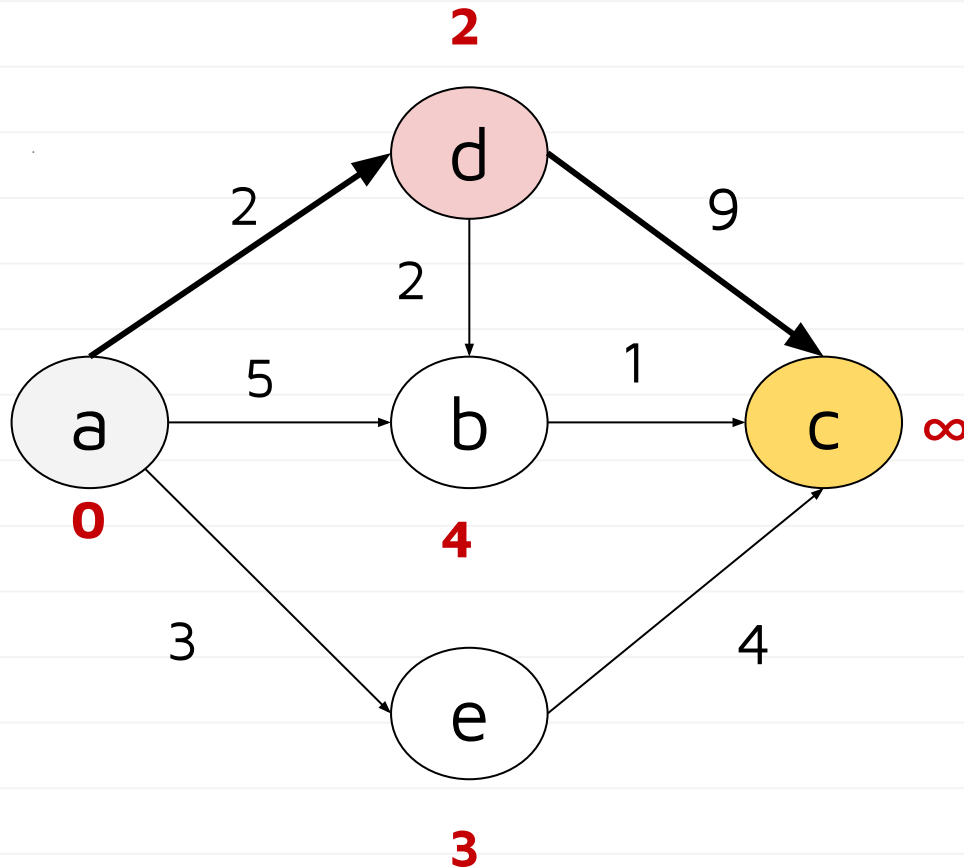
Select the **shortest** path: 2

Update rule:

If $2 + 2 < 5$, update

if $d(u) + \text{cost}(u, v) < d(v)$:
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Dijkstra's Algorithm



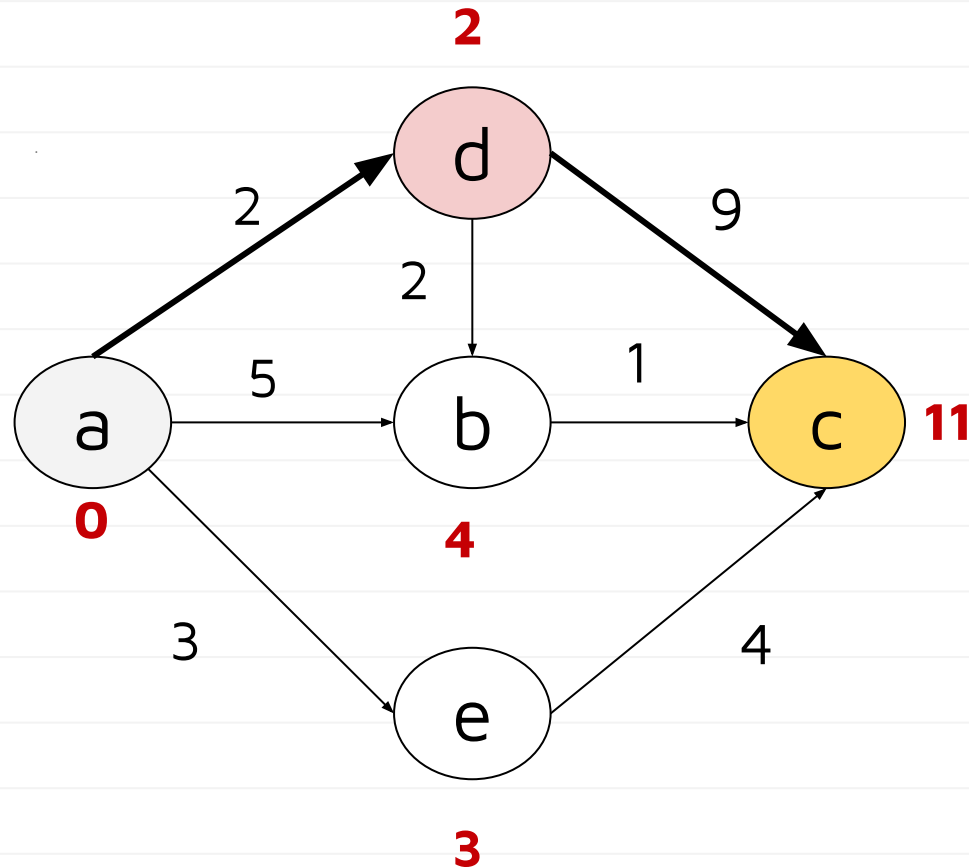
Select the **shortest** path: 2

Update rule:

If $2 + 9 < \text{inf}$, update

if $d(u) + \text{cost}(u, v) < d(v)$:
 $d(v) = d(u) + \text{cost}(u, v)$

Dijkstra's Algorithm



Select the **shortest** path: 2

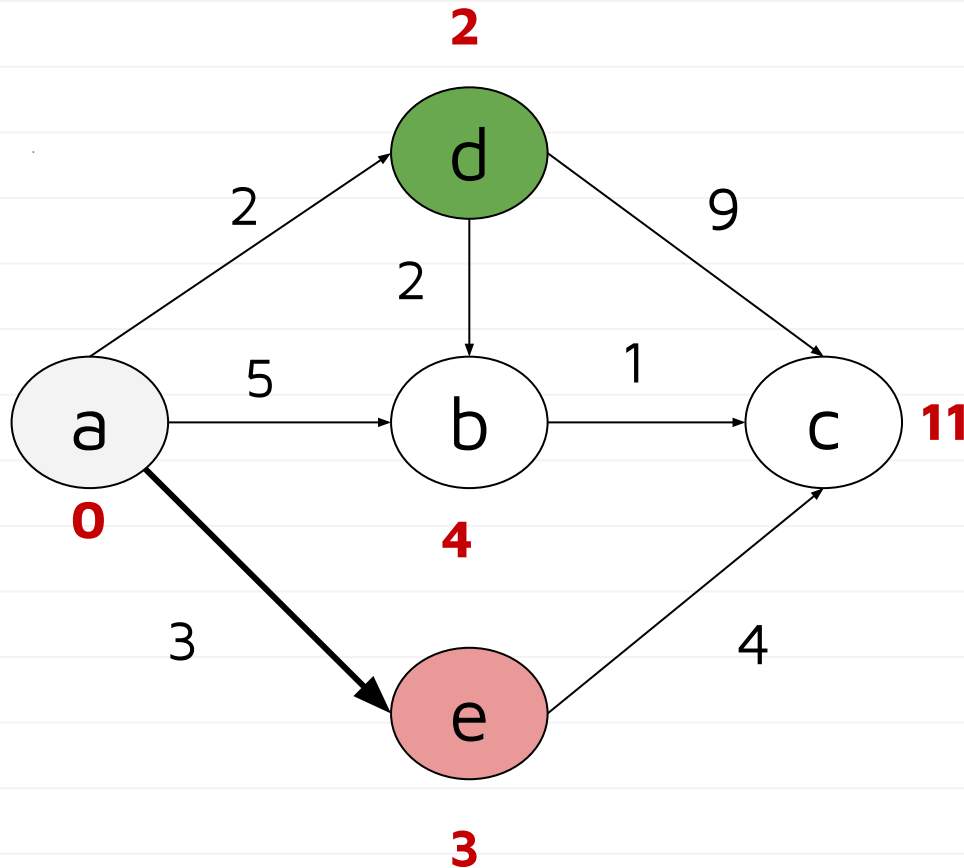
Update rule:

If $2 + 9 < \text{inf}$, update

if $d(u) + \text{cost}(u, v) < d(v)$:
 $d(v) = d(u) + \text{cost}(u, v)$

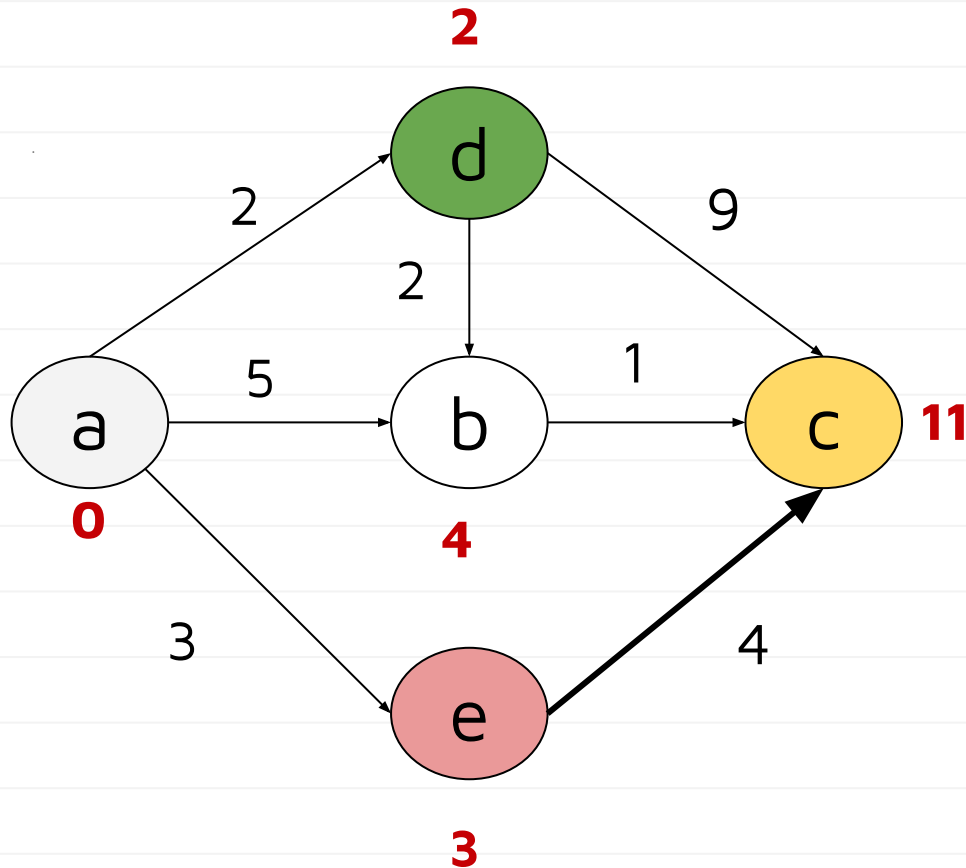
Dijkstra's Algorithm

Select the **shortest** path: 3



if $d(u) + \text{cost}(u, v) < d(v)$:
 $d(v) = d(u) + \text{cost}(u, v)$

Dijkstra's Algorithm



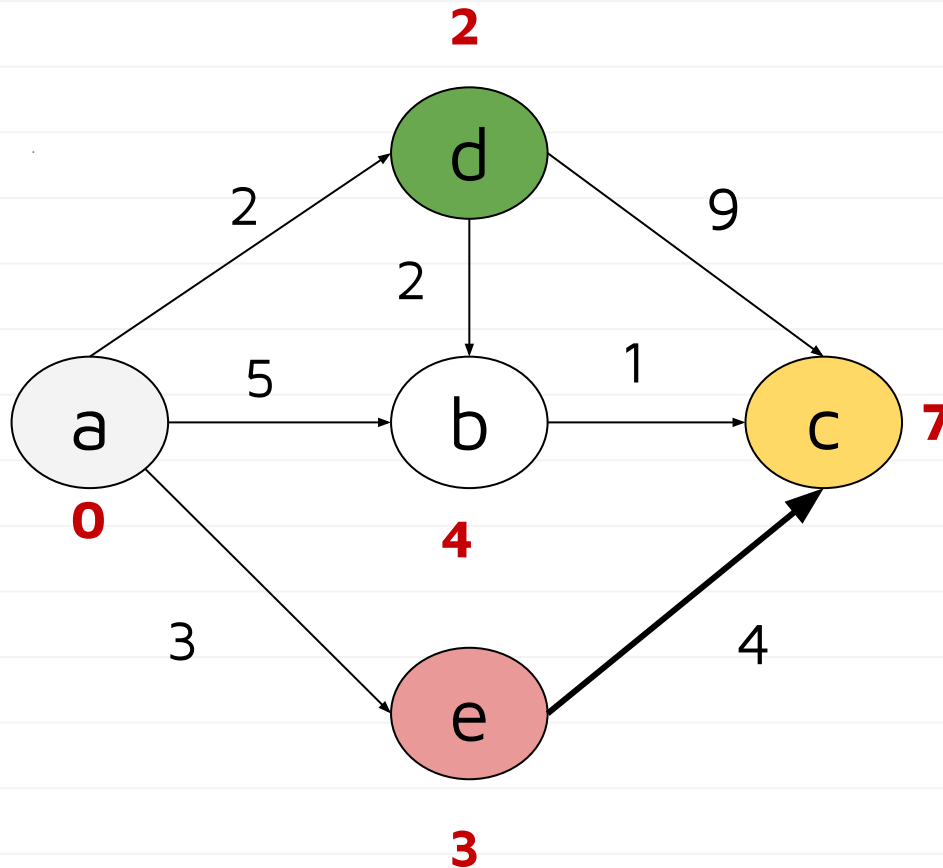
Select the **shortest** path: 3

Update rule:

If $3 + 4 < 11$, update

if $d(u) + \text{cost}(u, v) < d(v)$:
 $d(v) = d(u) + \text{cost}(u, v)$

Dijkstra's Algorithm



Select the **shortest** path: 3

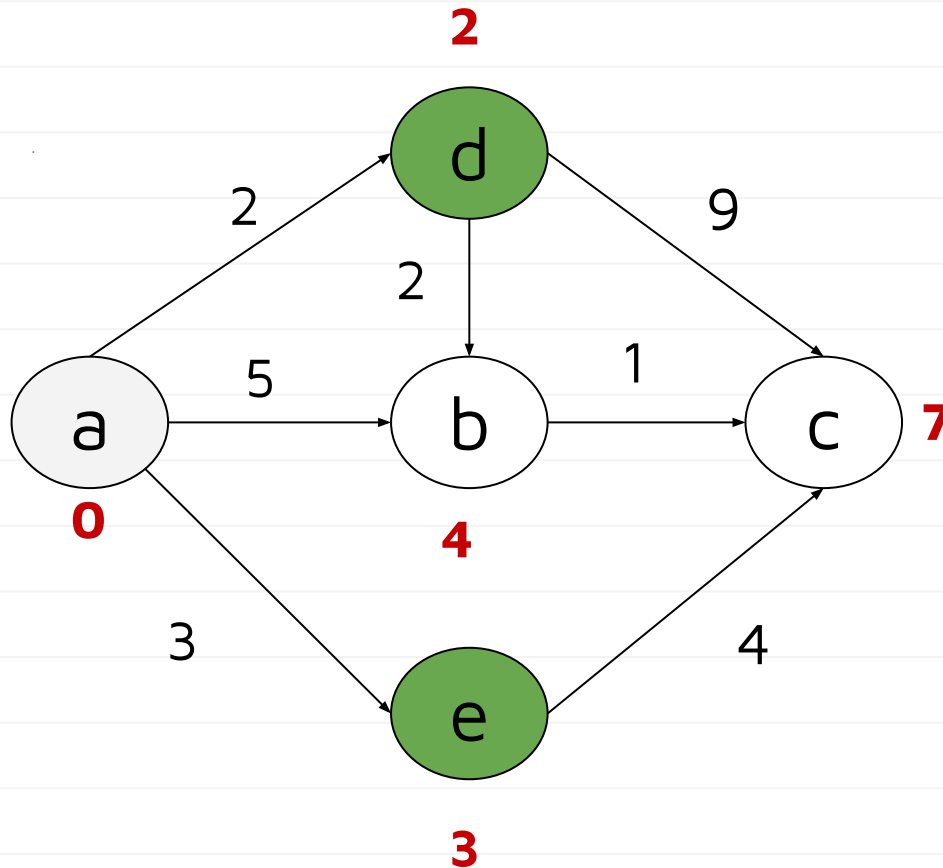
Update rule:

If $3 + 4 < 11$, update

if $d(u) + \text{cost}(u, v) < d(v)$:
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Dijkstra's Algorithm

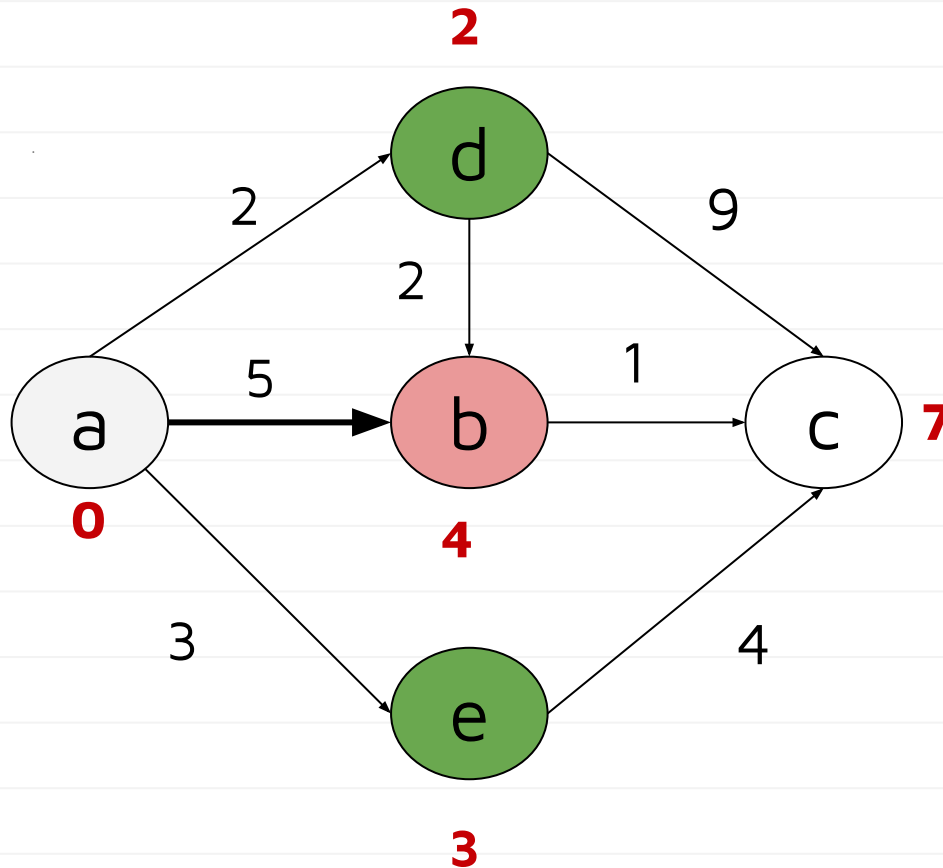
Select the **shortest** path: 4



if $d(u) + \text{cost}(u, v) < d(v)$:
 $d(v) = d(u) + \text{cost}(u, v)$

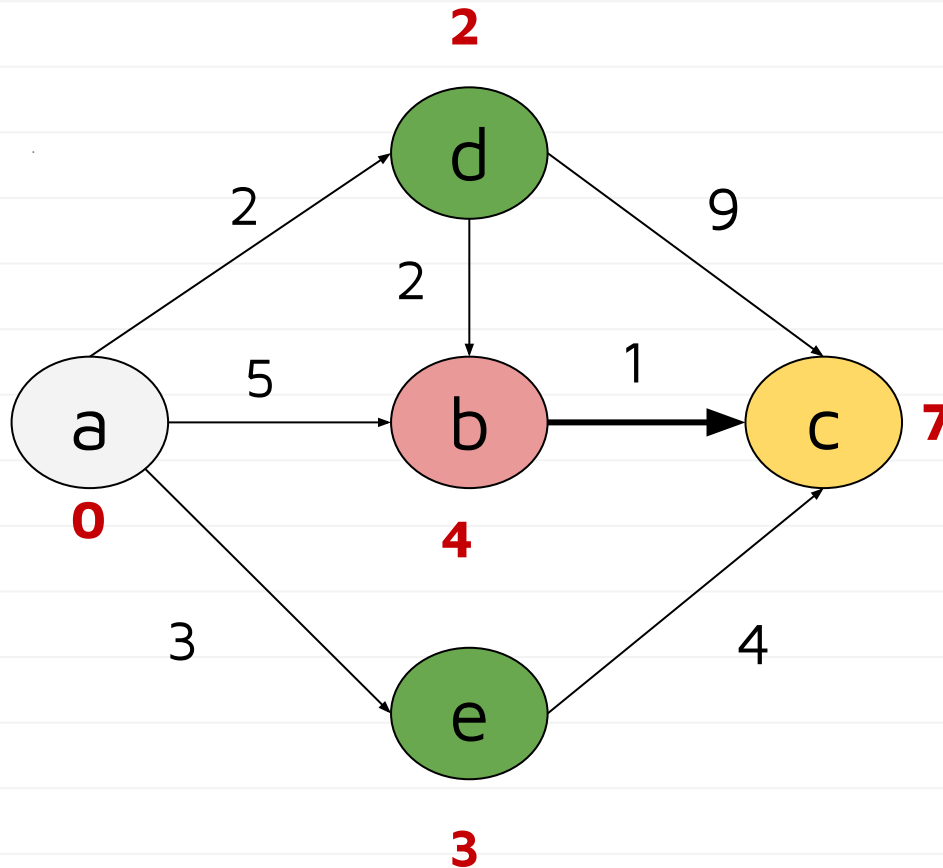
Dijkstra's Algorithm

Select the **shortest** path: 4



if $d(u) + \text{cost}(u, v) < d(v)$:
 $d(v) = d(u) + \text{cost}(u, v)$

Dijkstra's Algorithm



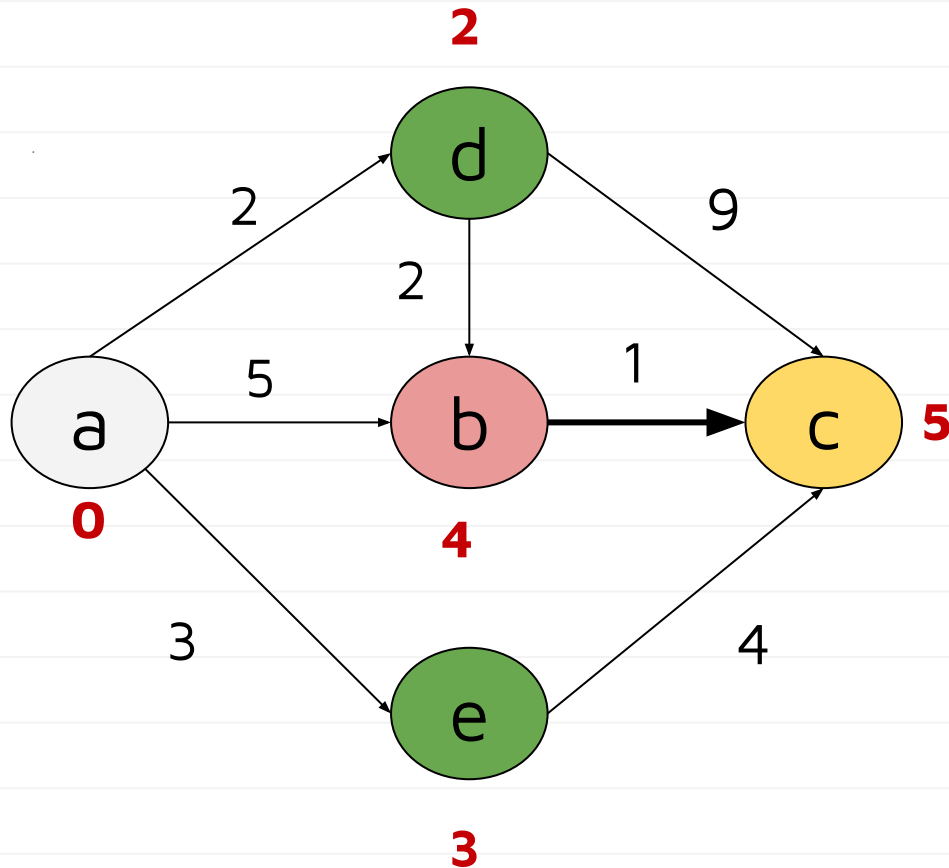
Select the **shortest** path: 4

Update rule:

If $4 + 1 < 7$, update

if $d(u) + \text{cost}(u, v) < d(v)$:
 $d(v) = d(u) + \text{cost}(u, v)$

Dijkstra's Algorithm



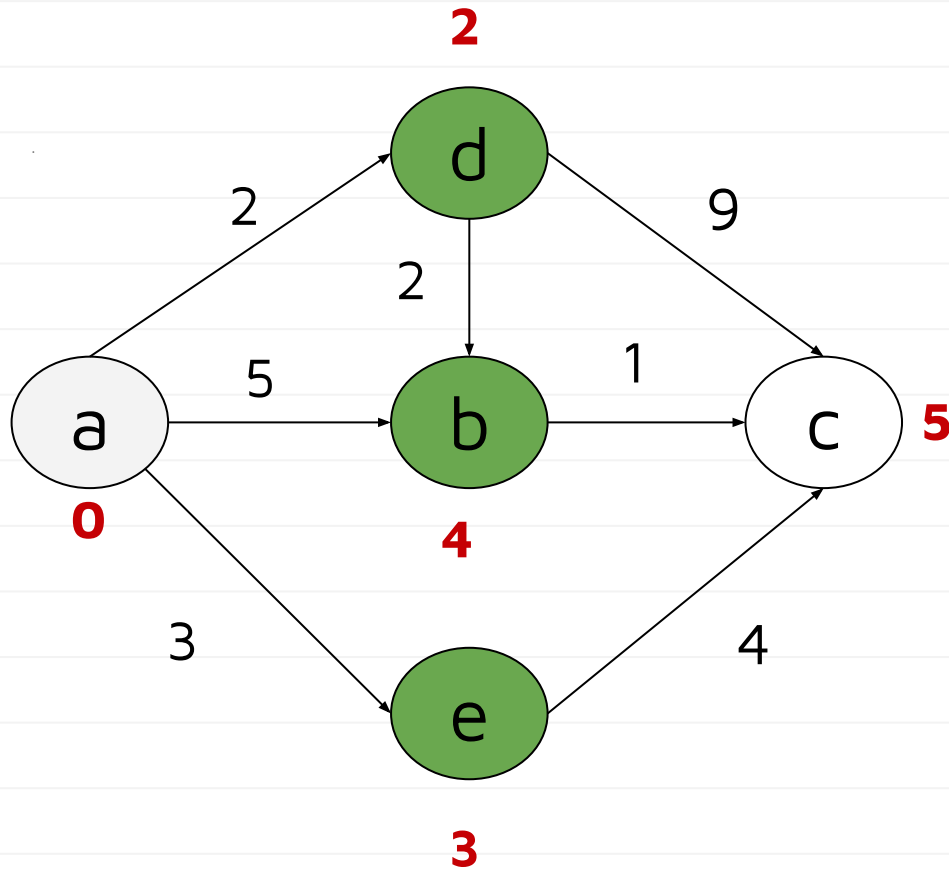
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Update rule:

If $4 + 1 < 7$, update

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Dijkstra's Algorithm



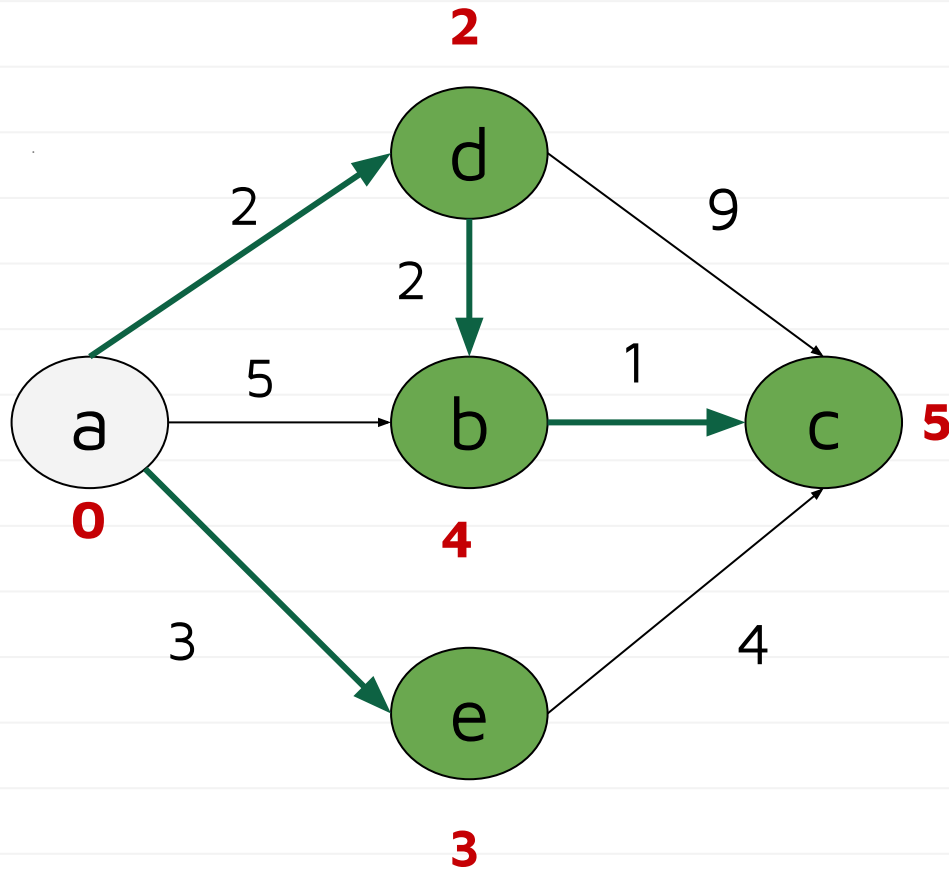
Select the **shortest** path: 4

Update rule:

If $4 + 1 < 7$, update

if $d(u) + \text{cost}(u, v) < d(v)$:
 $d(v) = d(u) + \text{cost}(u, v)$

Dijkstra's Algorithm



	a
b	4
c	5
d	2
e	3

if $d(u) + \text{cost}(u, v) < d(v)$:
 $d(v) = d(u) + \text{cost}(u, v)$

Dijkstra's Idea

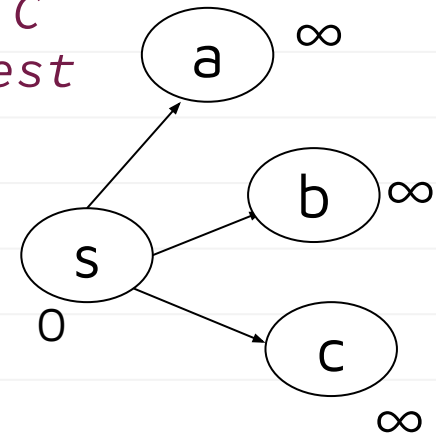
- Keep track of set C of “correct” nodes.
 - Nodes whose distance ***estimate*** is correct.
- At every step, add node *outside* of C with **smallest** estimated distance; update only its **neighbors**.
- A “**greedy**” algorithm.

Outline of Dijkstra's Algorithm

```
def dijkstra(graph, weights, source):  
    est = {node: float('inf') for node in graph.nodes}  
    est[source] = 0  
    pred = {node: None for node in graph.nodes}  
  
    # empty set  
    C = set()  
  
    # while there are nodes still outside of C  
    #     find node u outside of C with smallest  
    #     estimated distance  
    C.add(u)  
  
    for v in graph.neighbors(u):  
        update(u, v, weights, est, pred)  
  
    return est, pred
```

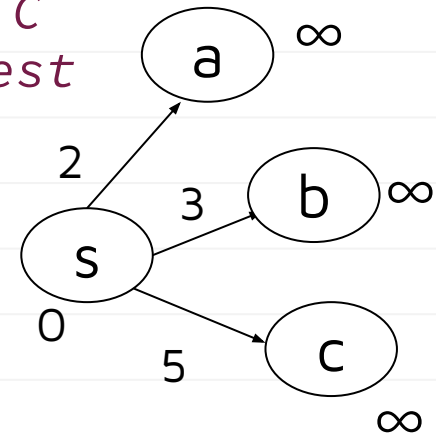
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    C = set()  
  
    # while there are nodes still outside of C  
    #   find node u outside of C with smallest  
    #   estimated distance  
    C.add(u)  
  
    for v in graph.neighbors(u):  
        update(u, v, weights, est, pred)  
  
    return est, pred
```



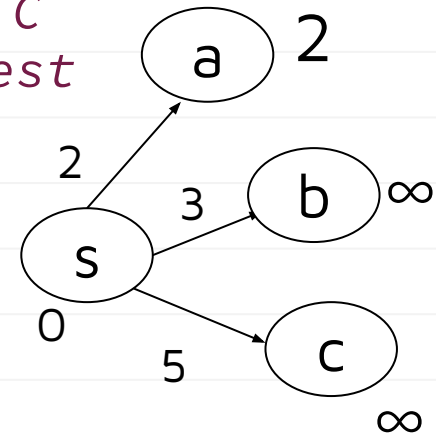
Outline of Dijkstra's Algorithm

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    # while there are nodes still outside of C  
    #     find node u outside of C with smallest  
    #     estimated distance  
    C.add(u)  
  
    for v in graph.neighbors(u):  
        update(u, v, weights, est, pred)  
  
    return est, pred
```



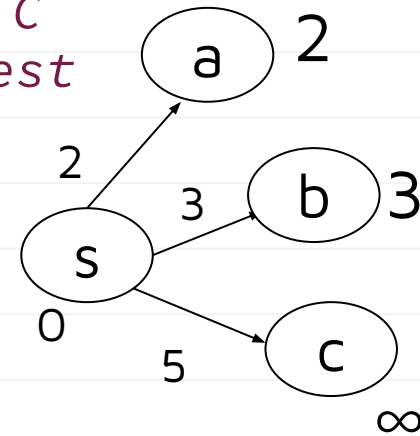
Outline of Dijkstra's Algorithm

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    # empty set  
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    # while there are nodes still outside of C  
    #   find node u outside of C with smallest  
    #   estimated distance  
    C.add(u)  
  
    for v in graph.neighbors(u):  
        update(u, v, weights, est, pred)  
  
    return est, pred
```



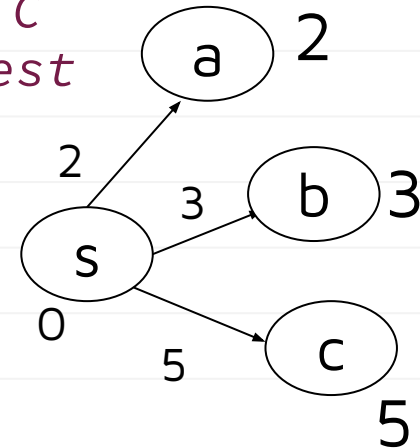
Outline of Dijkstra's Algorithm

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    # empty set  
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    # while there are nodes still outside of C  
    #     find node u outside of C with smallest  
    #     estimated distance  
    C.add(u)  
  
    for v in graph.neighbors(u):  
        update(u, v, weights, est, pred)  
  
    return est, pred
```



Outline of Dijkstra's Algorithm

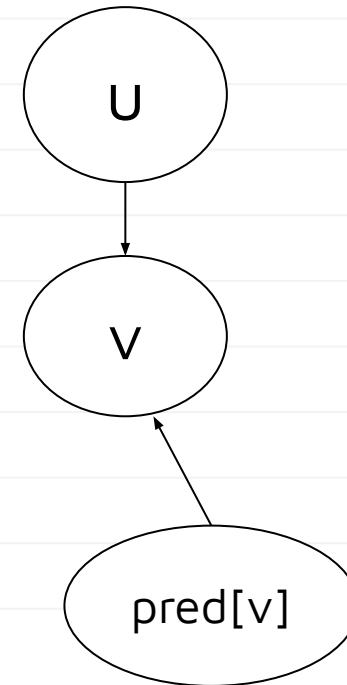
```
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    pred = {node: None for node in graph.nodes}  
  
    # empty set  
    C = set()  
  
    # while there are nodes still outside of C  
    #     find node u outside of C with smallest  
    #     estimated distance  
    C.add(u)  
  
    for v in graph.neighbors(u):  
        update(u, v, weights, est, pred)  
  
    return est, pred
```



Updating, in Code

```
def update(u, v, weights, est, predecessor):  
    """Update edge (u,v)."""  
    if est[v] > est[u] + weights(u,v):  
        est[v] = est[u] + weights(u,v)  
        predecessor[v] = u  
    return True  
else:  
    return False
```

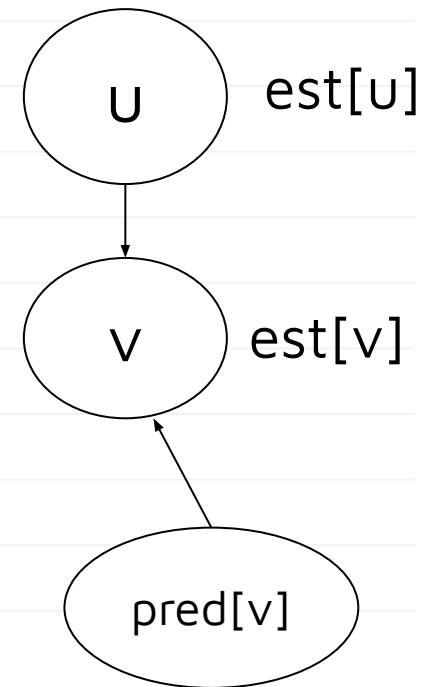
Time complexity: ?



Updating, in Code

```
def update(u, v, weights, est, predecessor):  
    """Update edge (u,v)."""  
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```

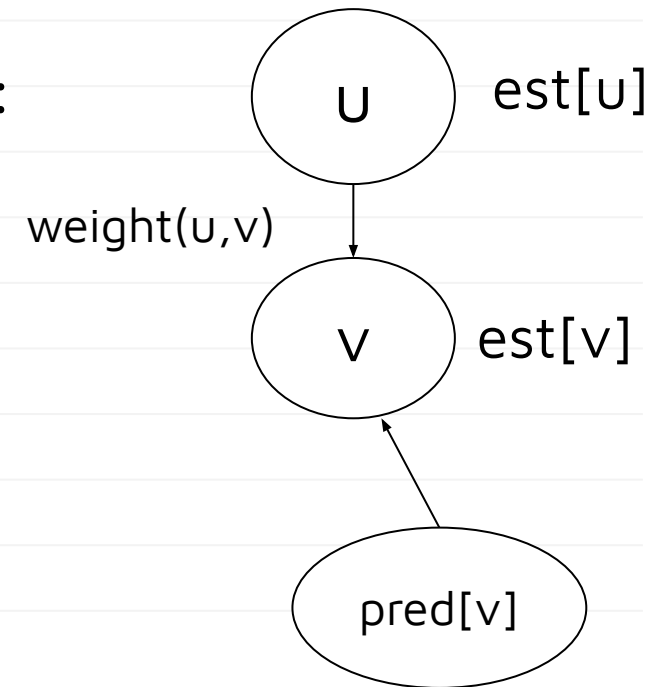
Time complexity: ?



Updating, in Code

```
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        predecessor[v] = u  
    return True  
else:  
    return False
```

Time complexity: ?



Updating, in Code

```
def update(u, v, weights, est, predecessor):  
    """Update edge (u,v)."""  
    if est[v] > est[u] + weights(u,v):  
        est[v] = est[u] + weights(u,v)  
        predecessor[v] = u  
    return True  
else:  
    return False
```

Time complexity: ?

A: $\Theta(1)$

B: $\Theta(E)$

C: $\Theta(V)$

D: $\Theta(V + E)$

Updating, in Code

```
def update(u, v, weights, est, predecessor):  
    """Update edge (u,v)."""  
    if est[v] > est[u] + weights(u,v):  
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        predecessor[v] = u  
        return True  
    else:  
        return False
```

Time complexity: $\Theta(1)$

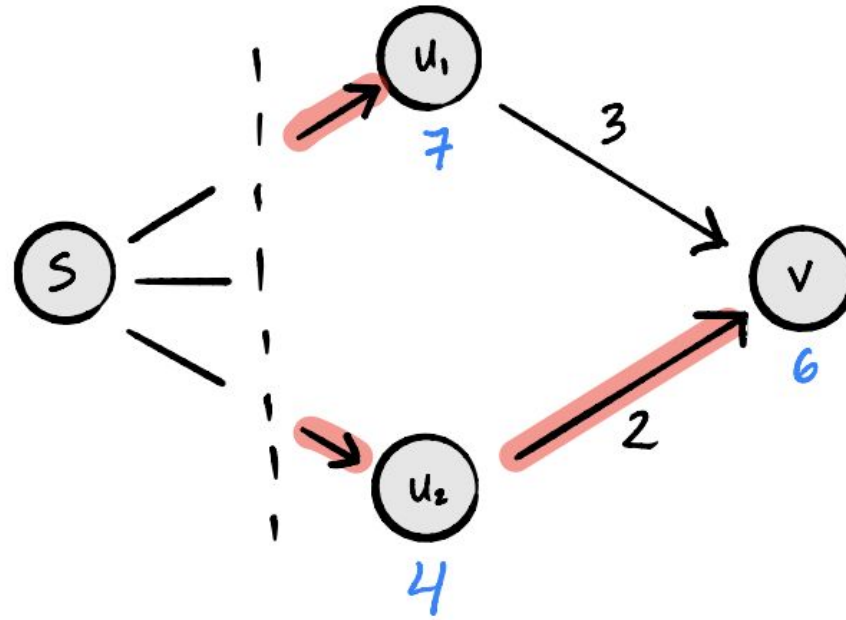
When does an update discover a shortest path?

Suppose updating (u_2, v) finds a shorter path to v .

True or **False**: the actual shortest path must go through u_2 .

A: True

B: False

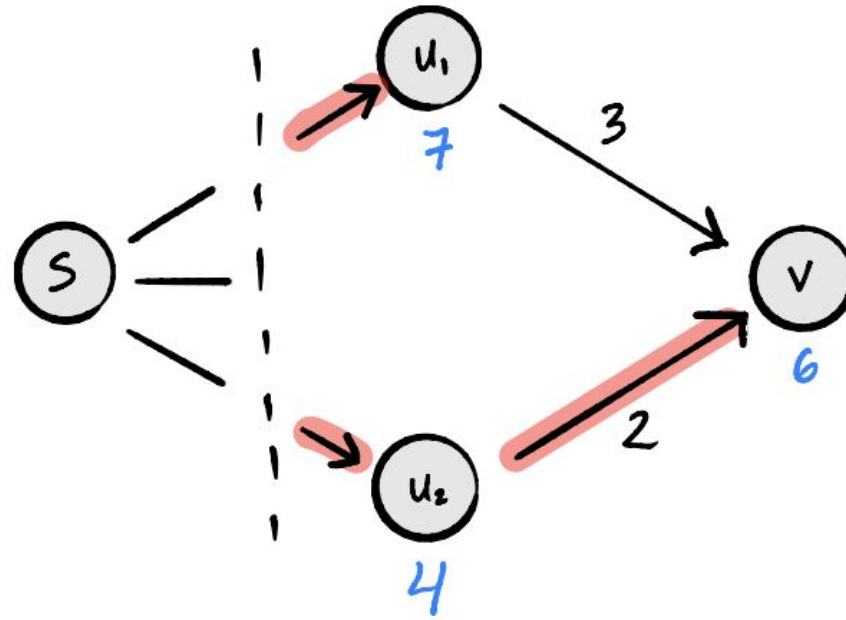


When does an update discover a shortest path?

Suppose updating (u_2, v) finds a shorter path to v .

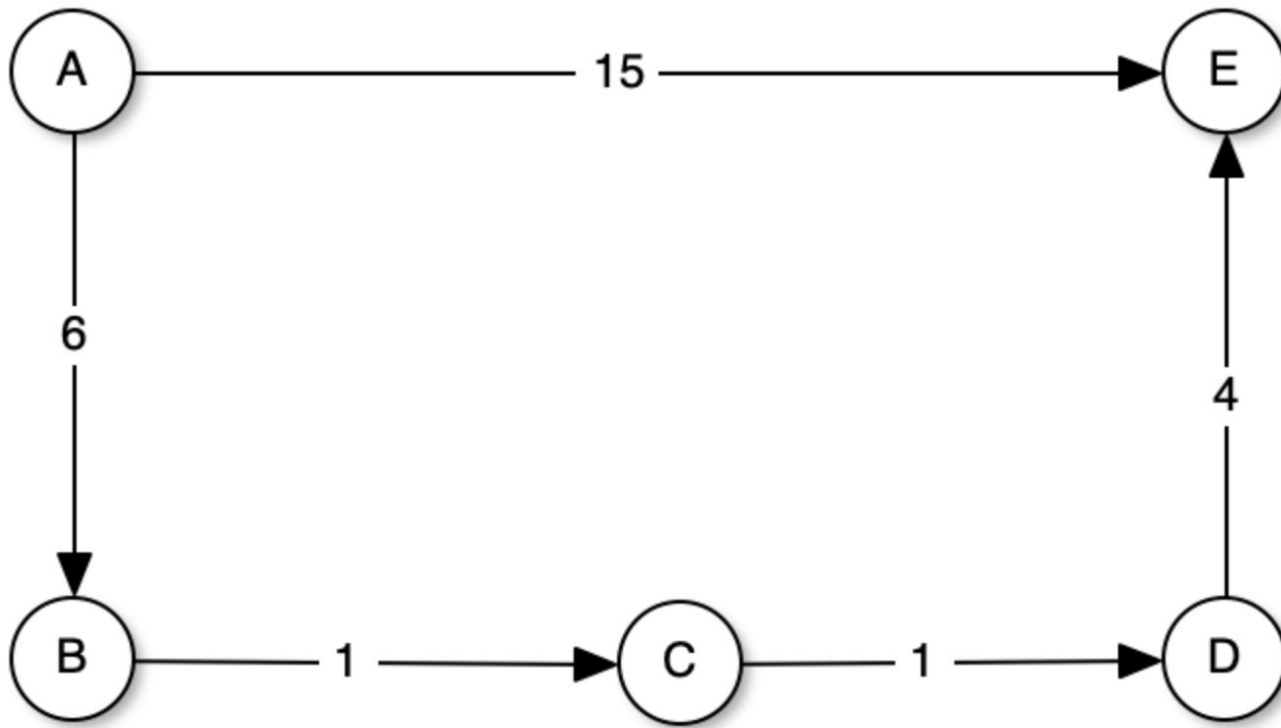
True or **False**: the actual shortest path must go through u_2 .

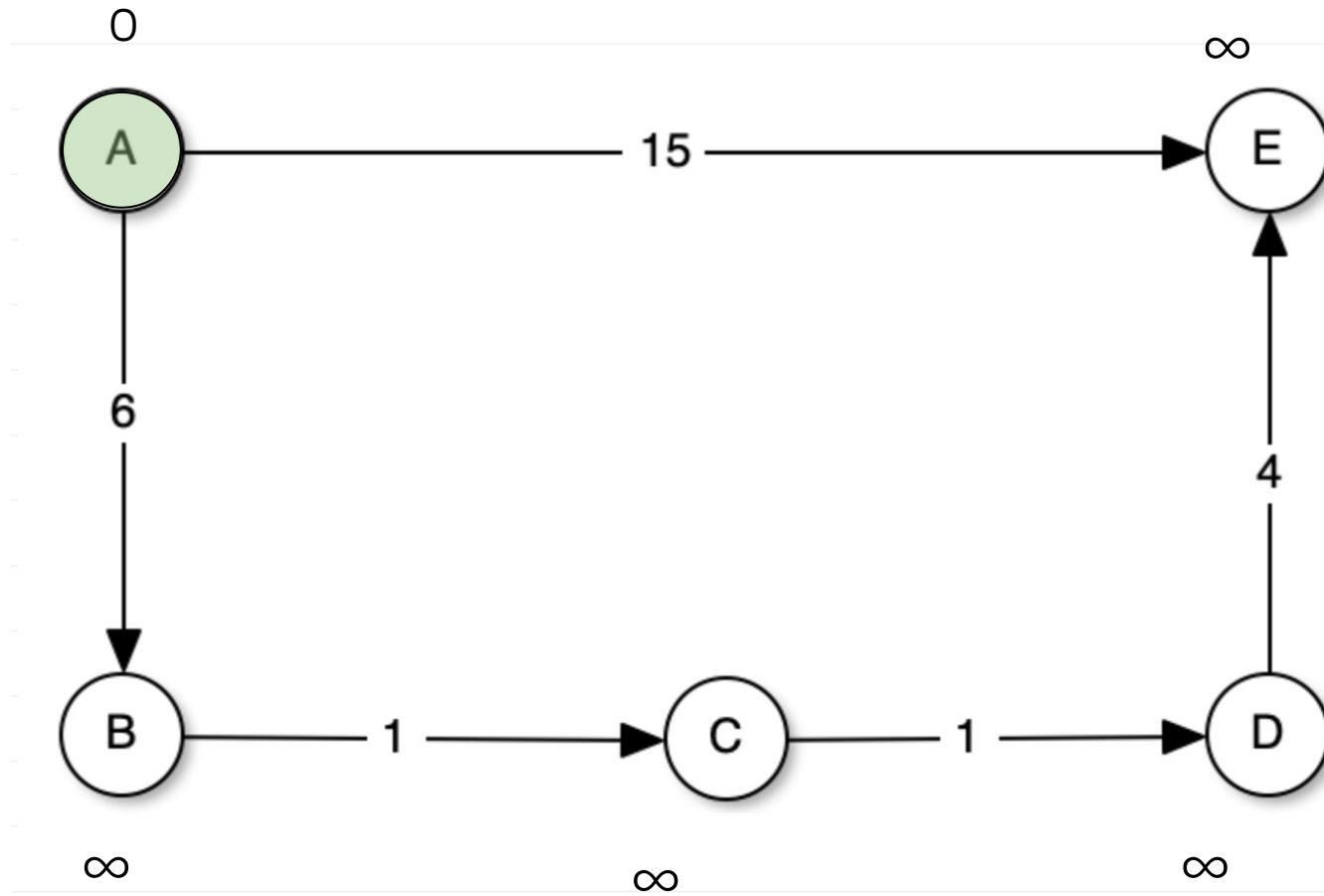
False: we might later discover a better path to u_1



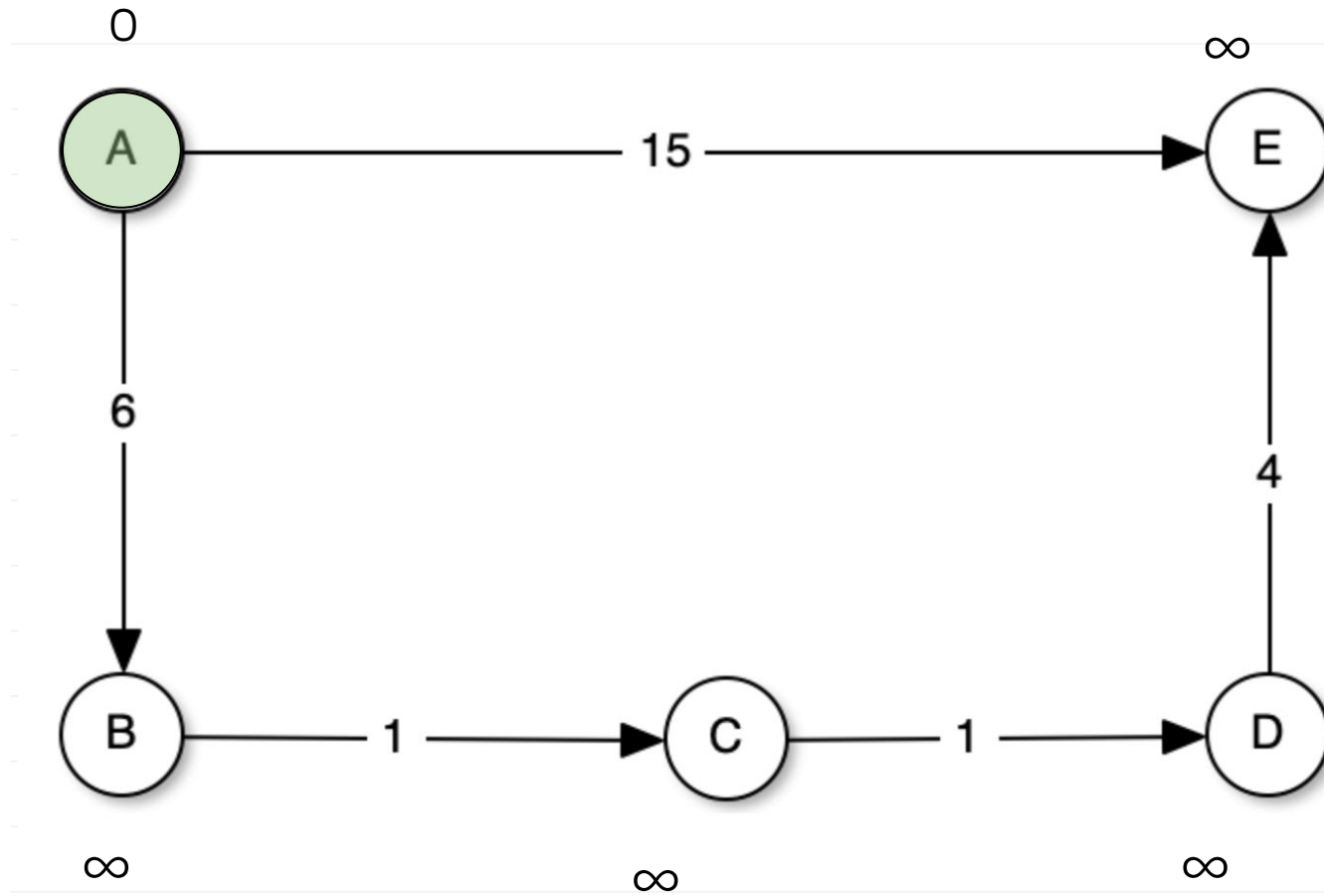
When does an update discover a shortest path?

- Let (u, v) be an edge we update.
- Suppose:
 - the **actual** shortest path to u has been found;
 - the **actual** shortest path to v goes through (u, v) .
- Then after *updating* (u, v) , the estimated shortest path to v is **correct**.

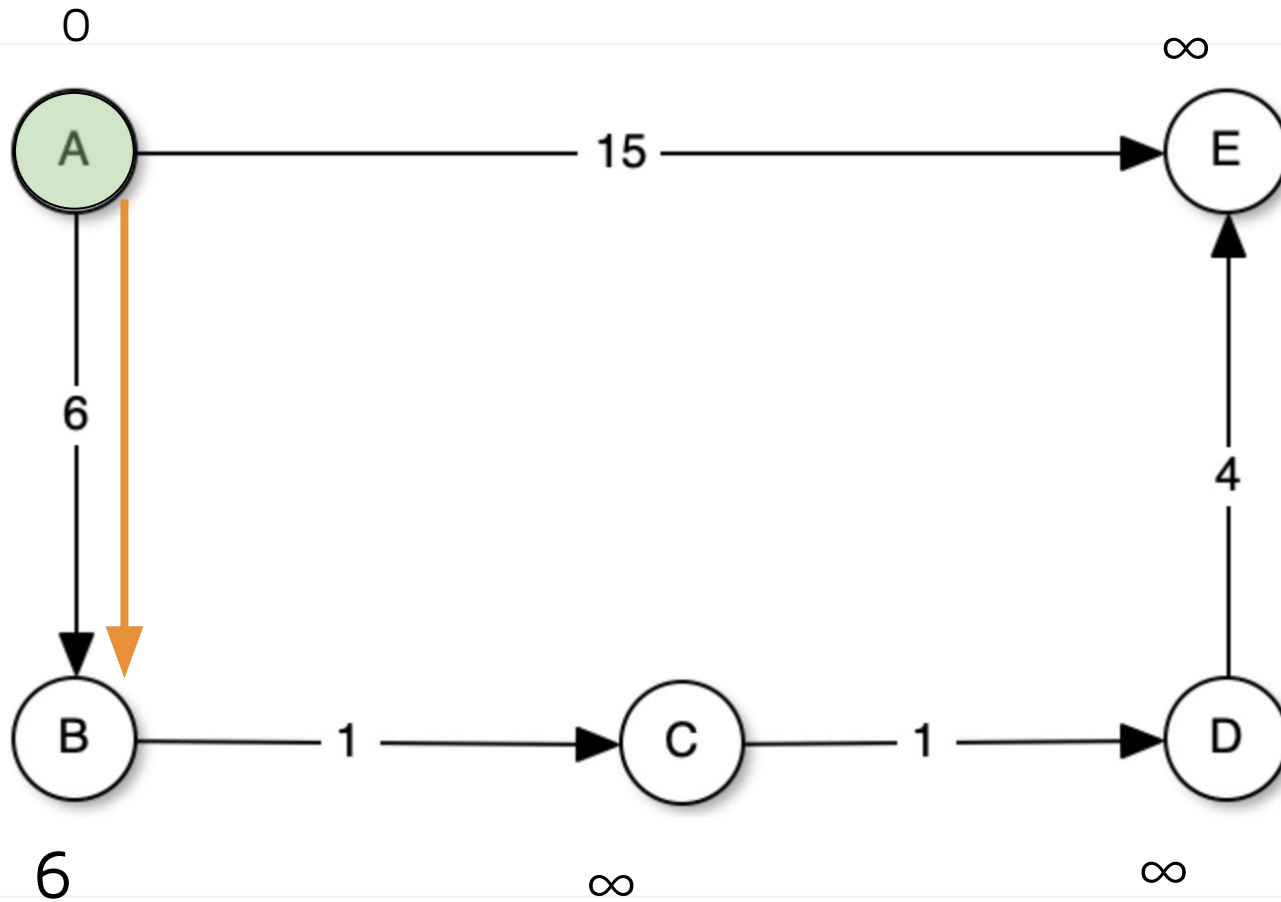




$C = \{ \}$

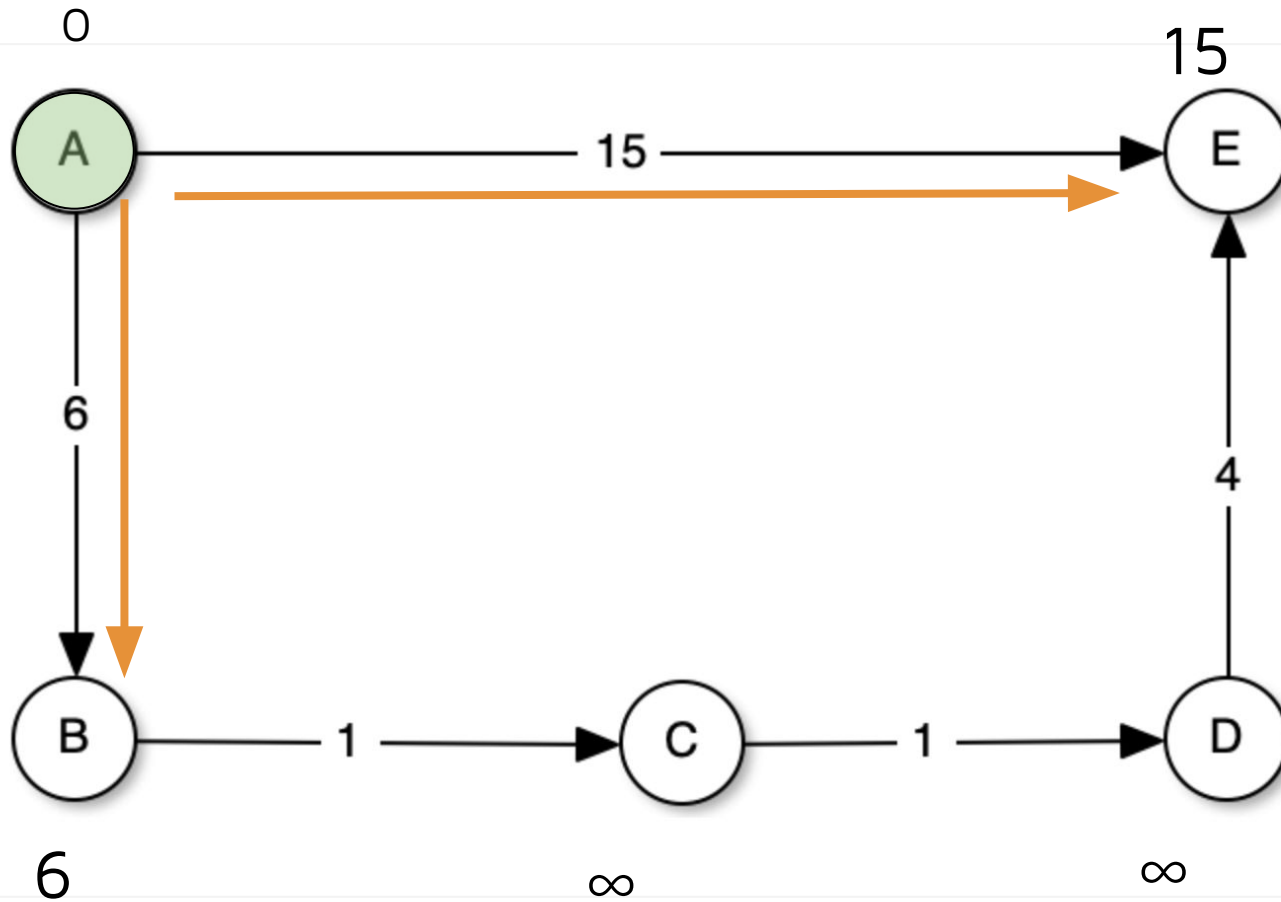


$C = \{ A \}$



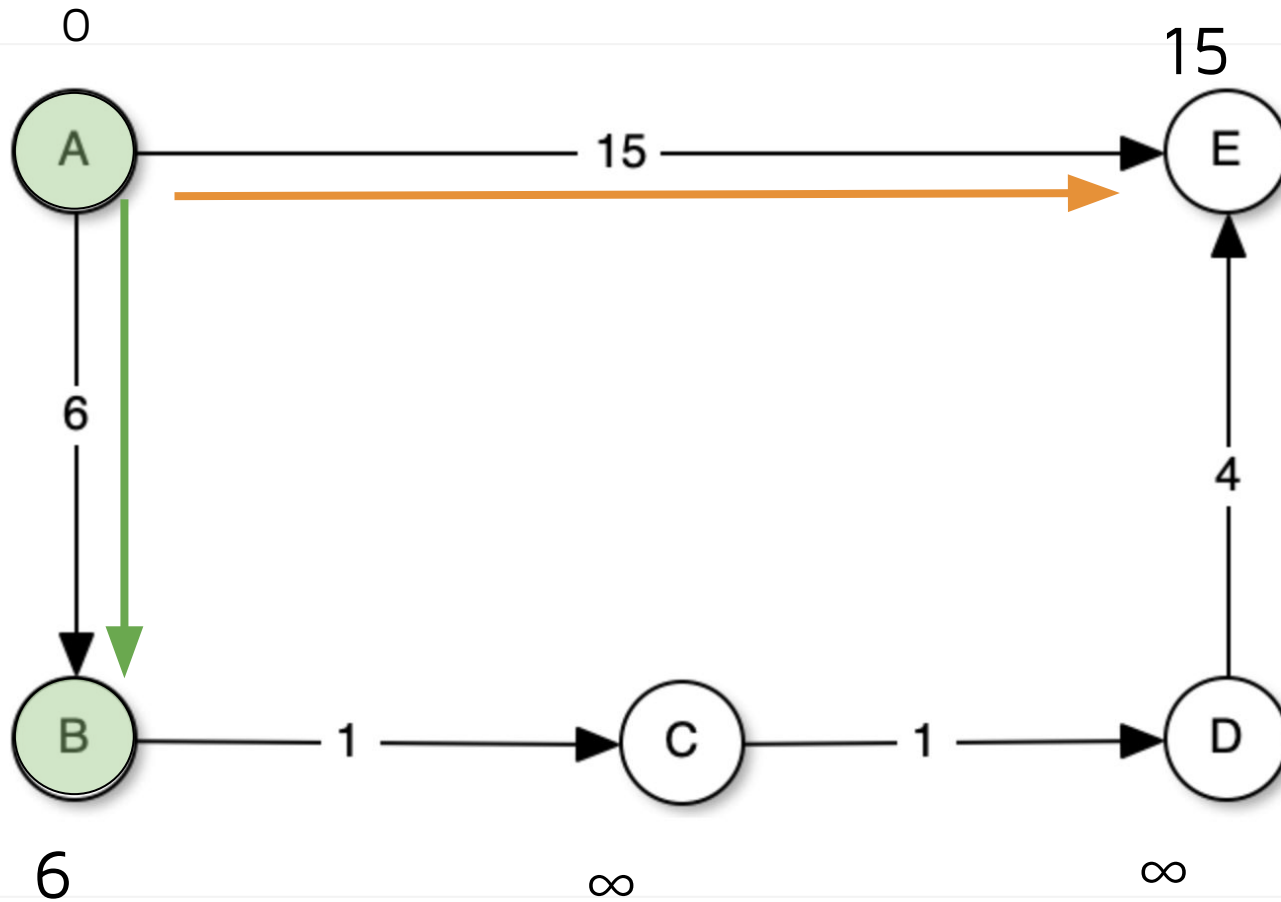
$C = \{ A \}$

pred =
 $\{ B:A,$



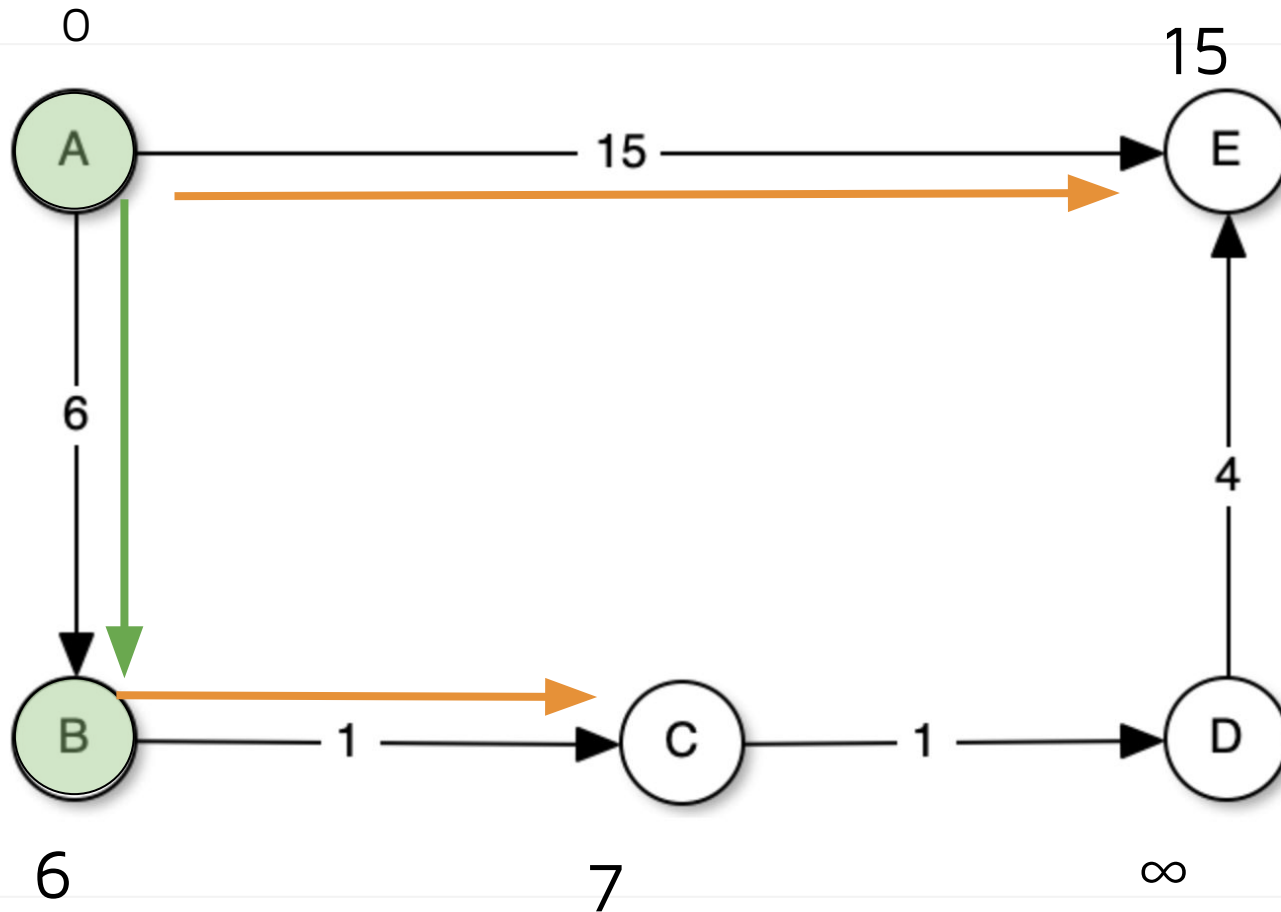
$C = \{ A \}$

pred =
{ B:A,
E:A, #for now



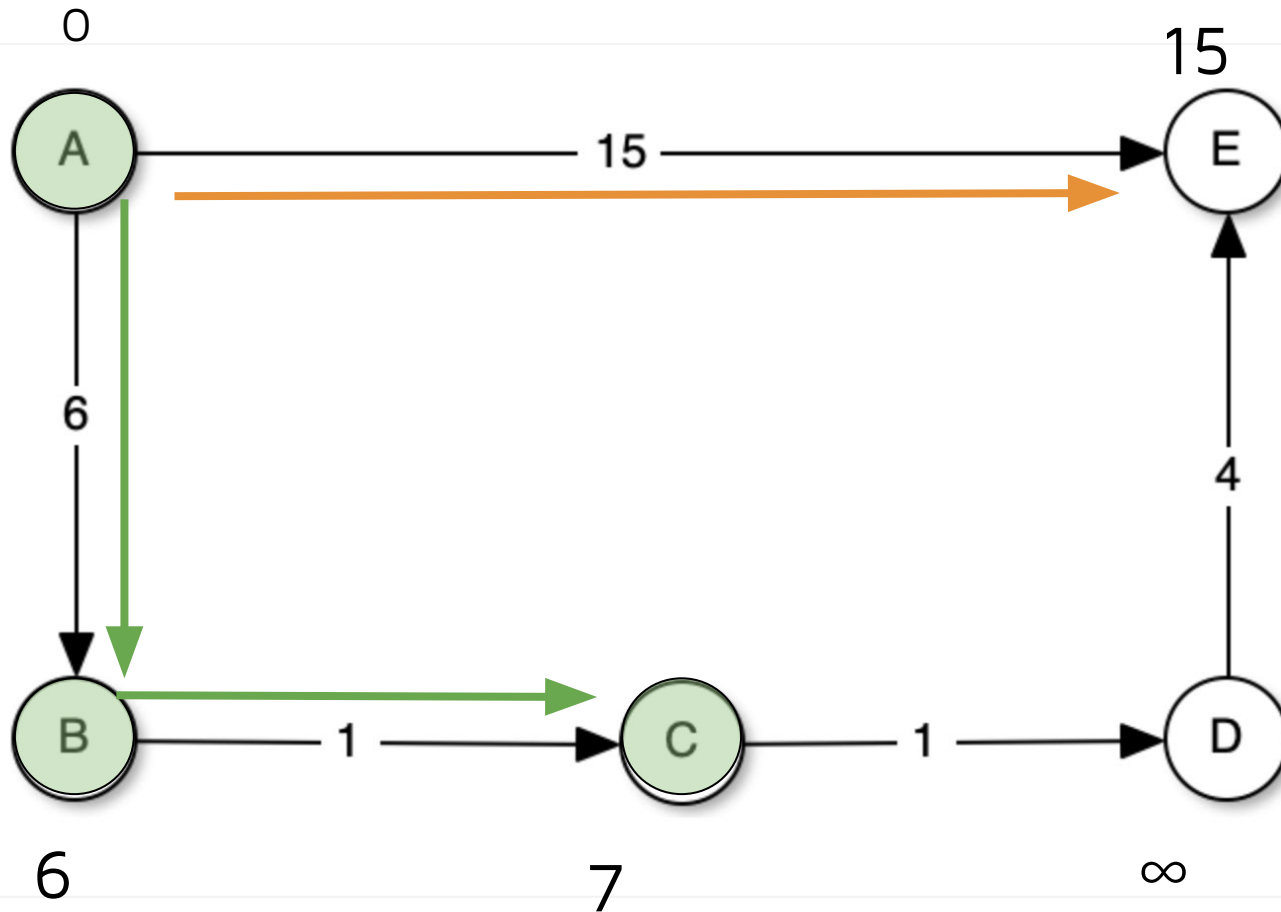
$C = \{ A, B \}$

pred =
 { B:A,
 E:A, #for now



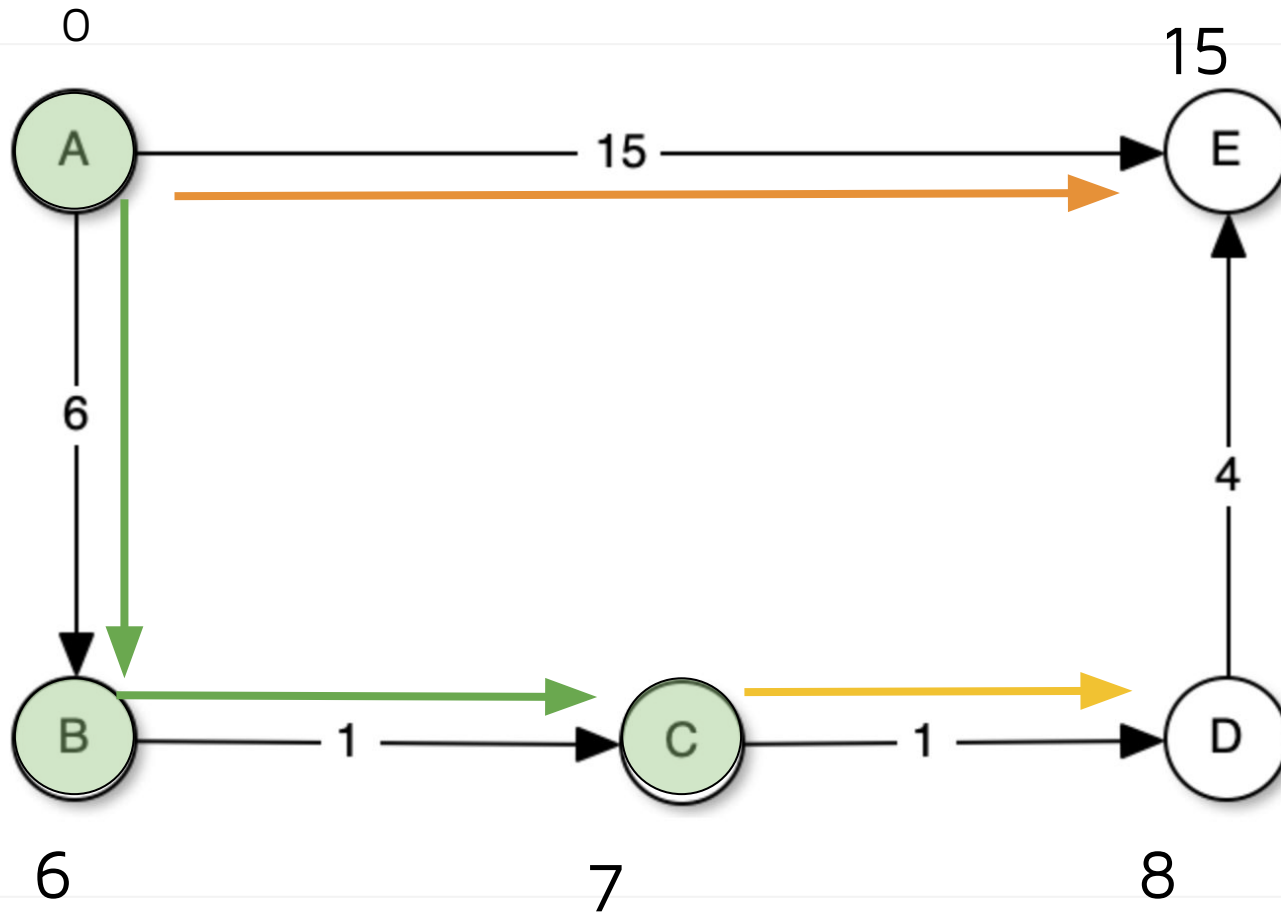
$C = \{ A, B \}$

pred =
{ B:A,
E:A, #for now
C:B,



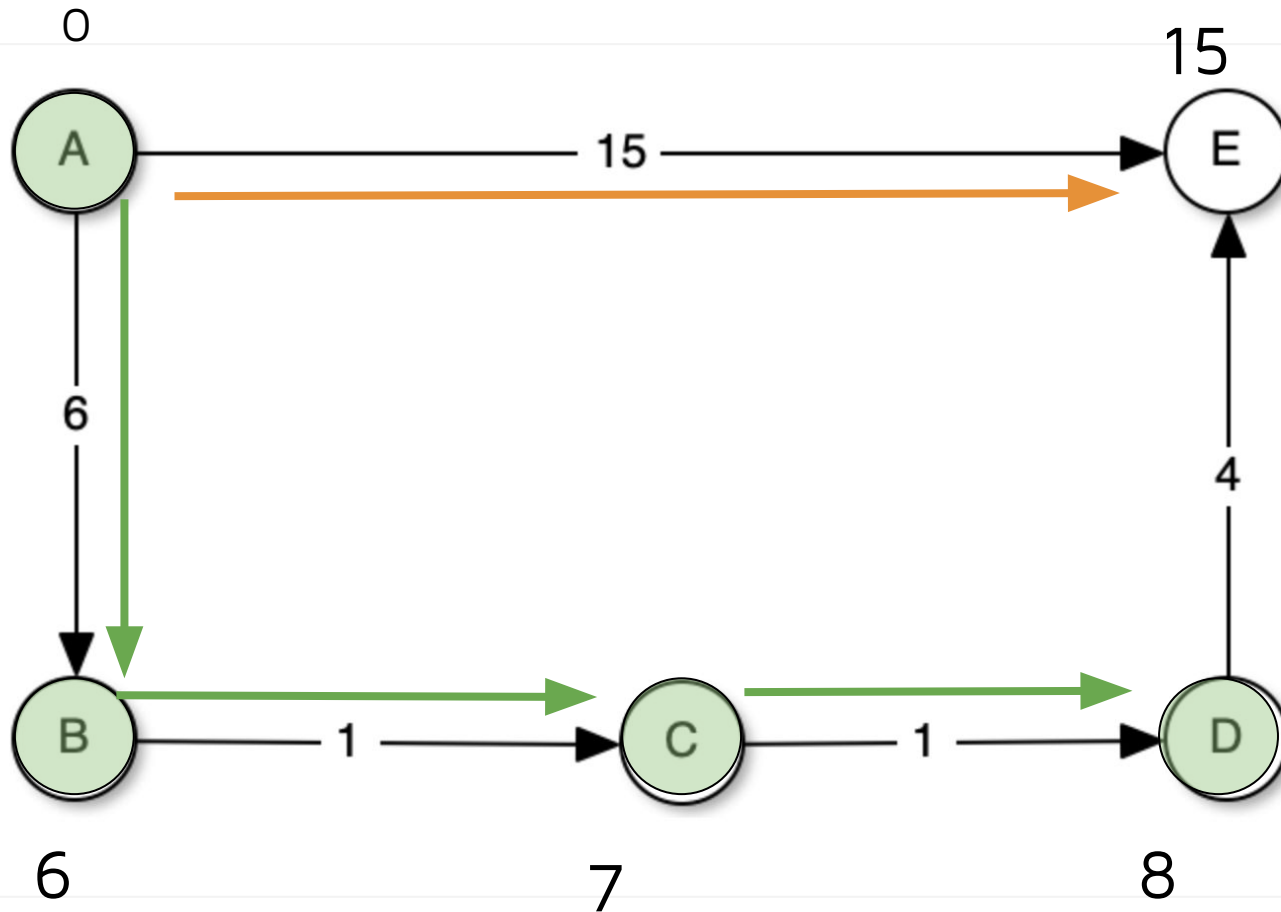
$C = \{ A, B, C \}$

pred =
{ B:A,
E:A, #for now
C:B,



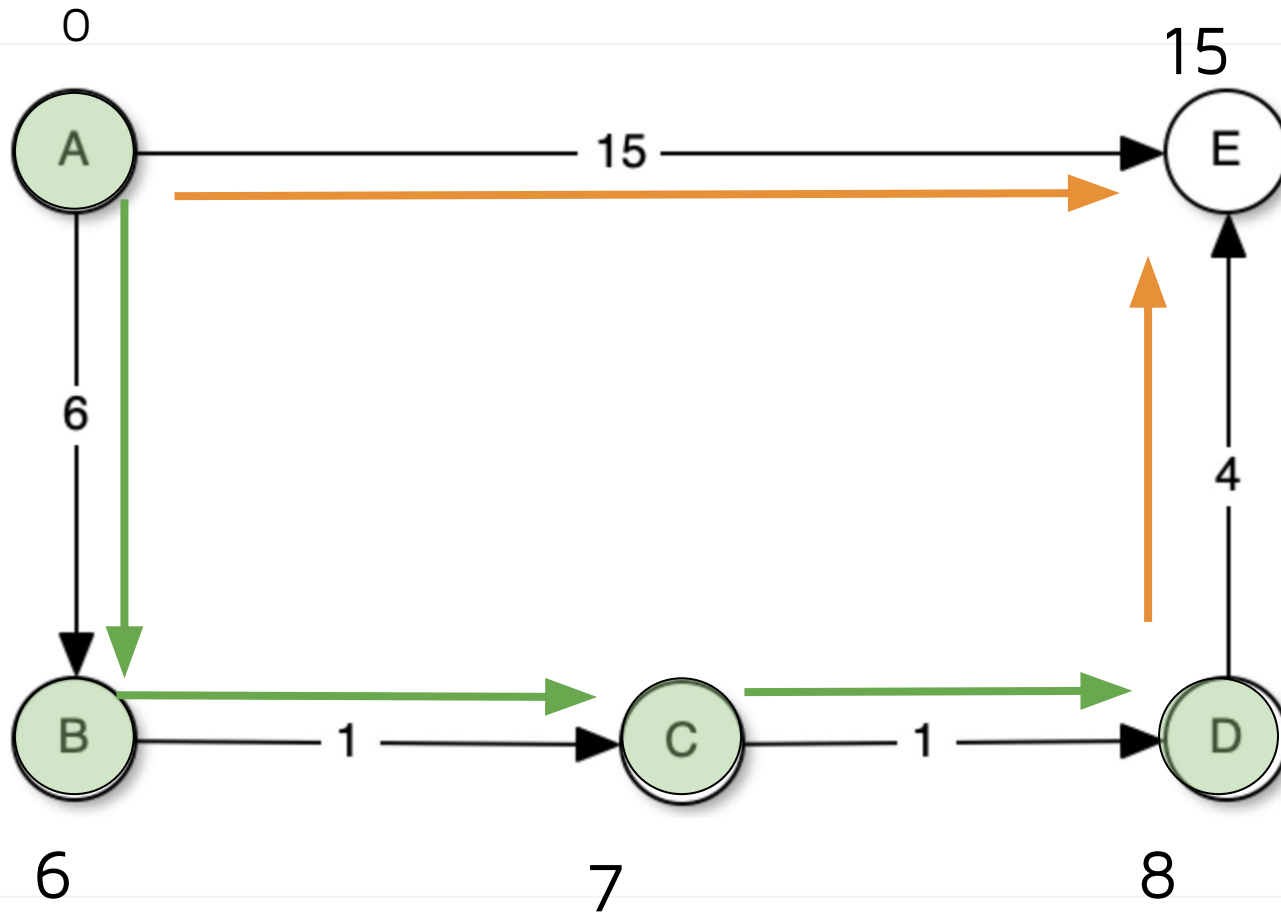
$C = \{ A, B, C \}$

pred =
{ B:A,
E:A, #for now
C:B,
D:C



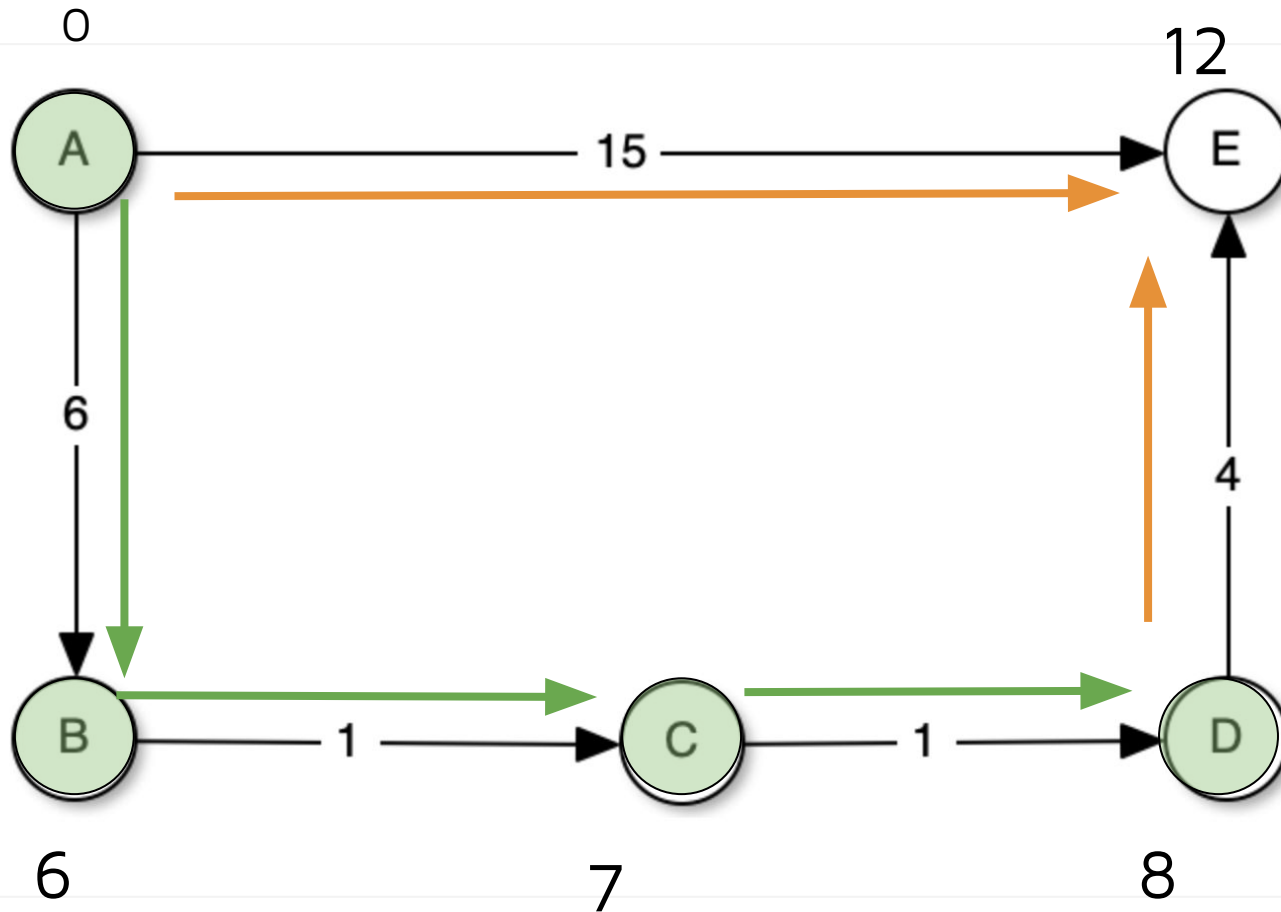
$C = \{ A, B, C, D \}$

pred =
{ B:A,
E:A, *#fornow*
C:B,
D:C



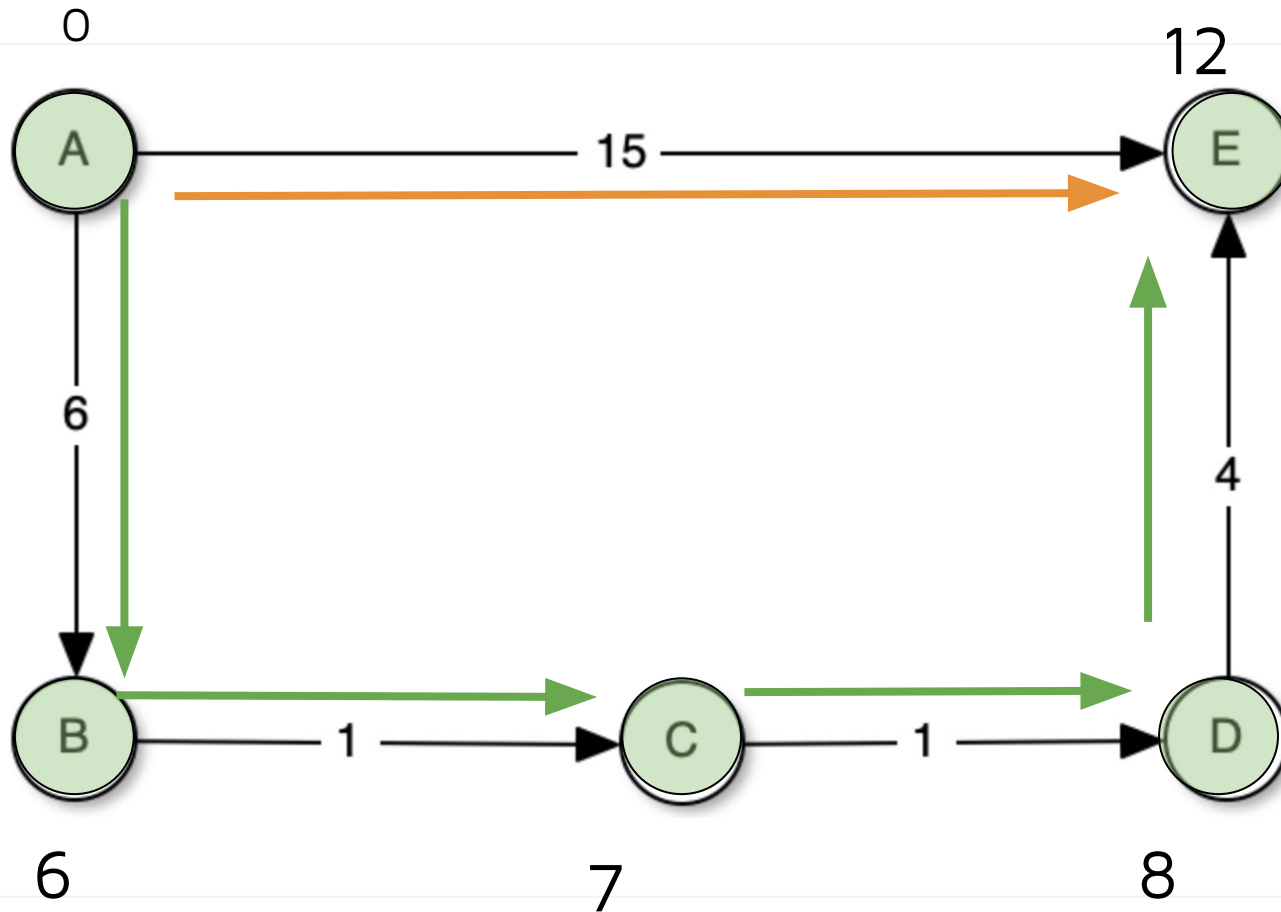
$C = \{ A, B, C, D \}$

pred =
{ B:A,
E:A, #for now
C:B,
D:C



$C = \{ A, B, C, D \}$

pred =
{ B:A,
E:D,
C:B,
D:C



How many times do we update ("touch") each edge?

A: Once

B: Twice

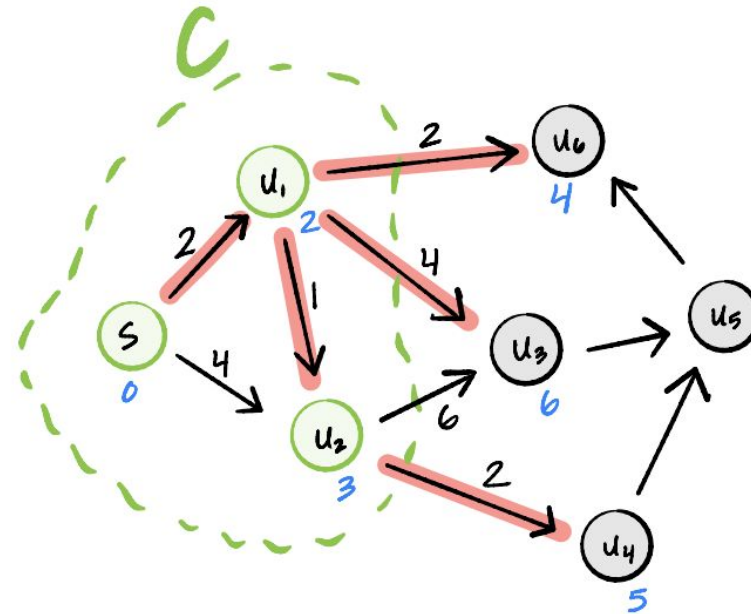
C: $|E|$

D: $|V|$

E: Something else

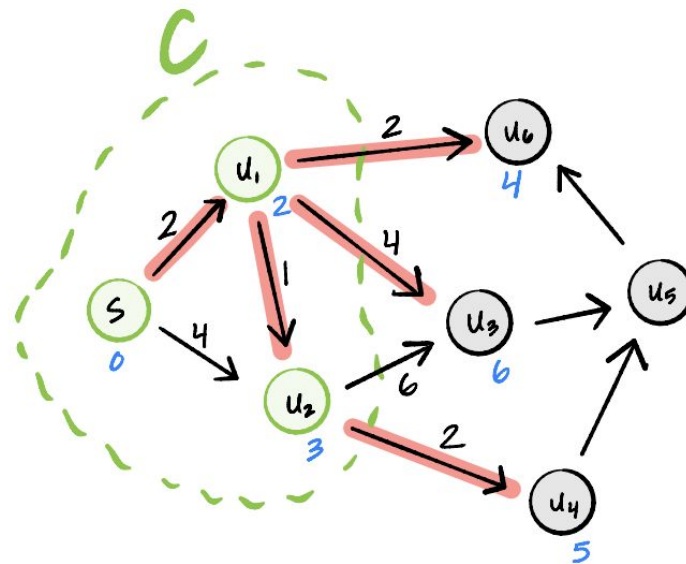
Proof Idea

Claim: at beginning of any iteration of Dijkstra's, if u is node $\notin C$ with smallest estimated distance, the shortest path to u has been correctly discovered.



Proof Idea

- Let u be node outside of C for which $\text{est}[u]$ is smallest.
- We've discovered a path from s to u of length $\text{est}[u]$.
- Any path from s to u has to exit C somewhere.
- Any path from s to u will cost at least $\text{est}[u]$ just to exit C .



Question

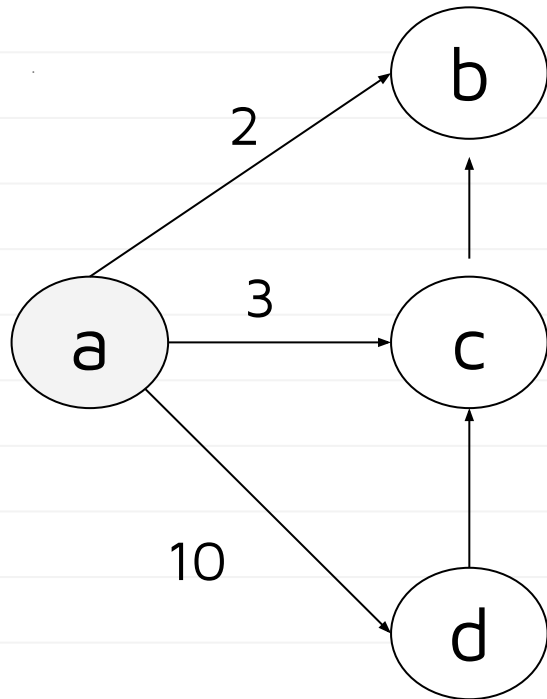
- Is there a limitation for Dijkstra's algorithm?

Exercise

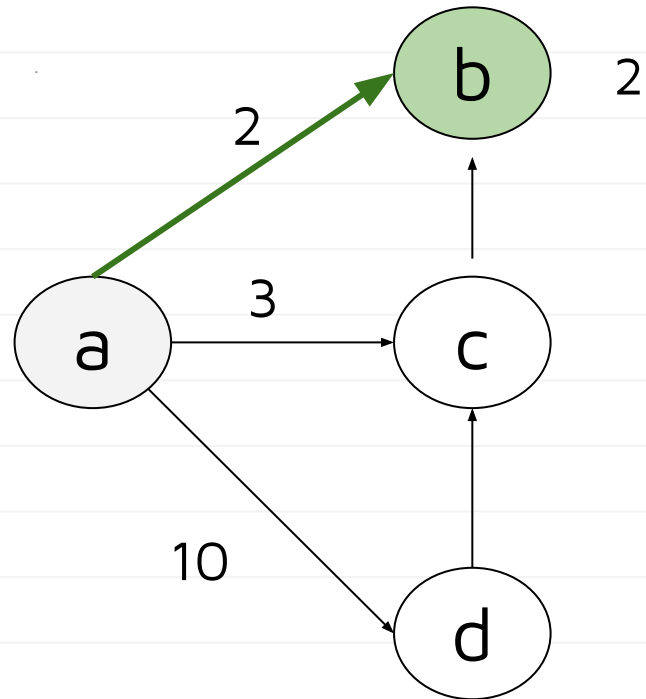
- Why do the edge weights need to be positive?
- Come up with a simple example graph with some negative edge weights where Dijkstra's **fails** to compute the correct shortest path.

Example

What is the fastest way to get to **b**?

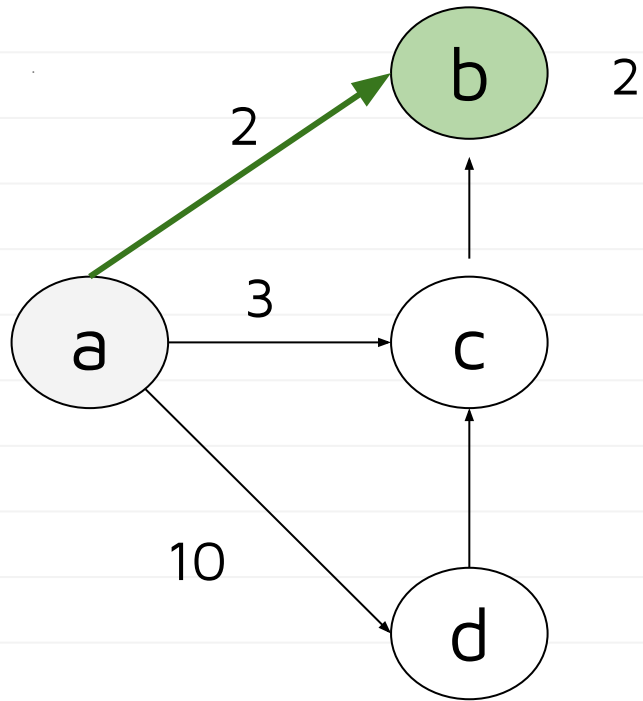


Example



What is the fastest way to get to **b**?

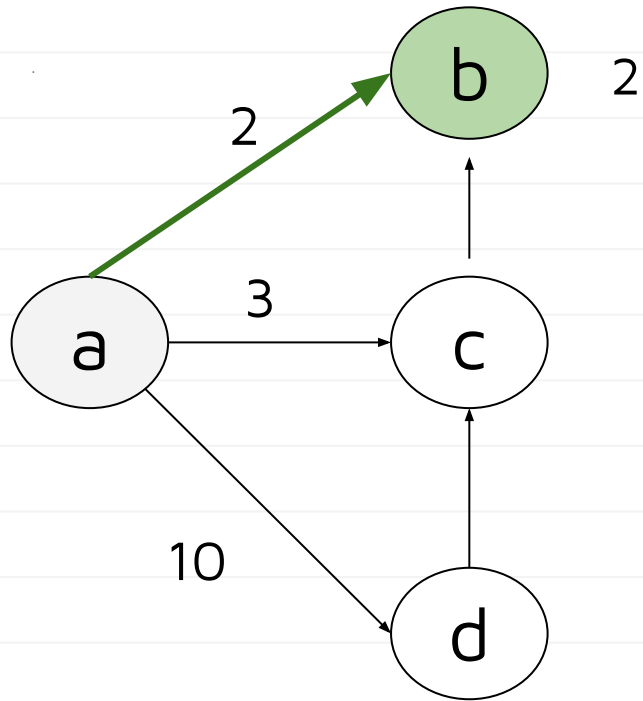
Example



What is the fastest way to get to **b**?

Do we ever have to consider another path to **b**?

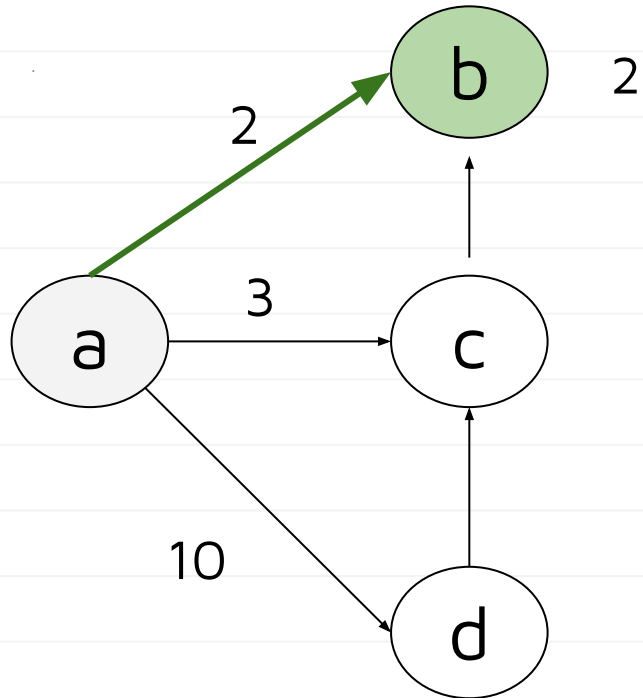
Example



What is the fastest way to get to **b**?

Do we ever have to consider another path to **b**? No.

Example

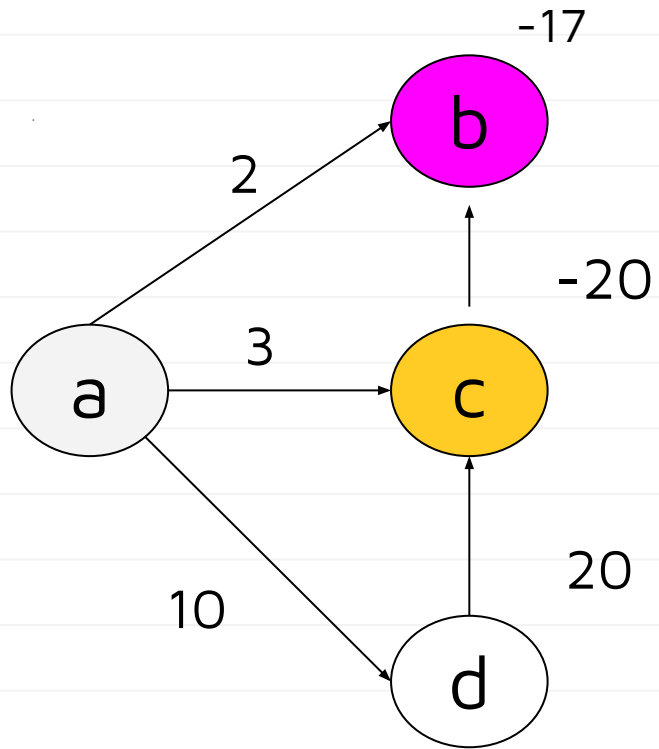


What is the fastest way to get to **b**?

Do we ever have to consider another path to **b**? No.

Let's make another path that is **better** than 2.

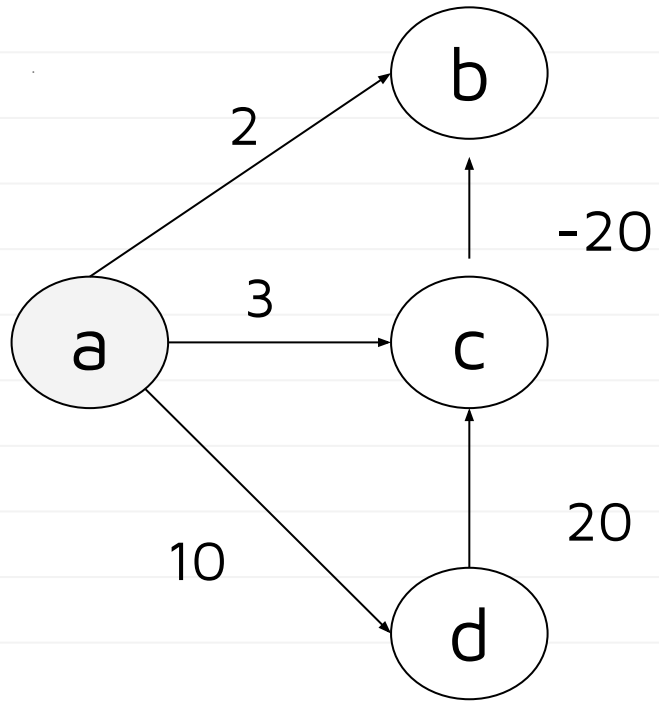
Example



Let's make another path that is **better** than 2.

Will the algorithm work for these weights?

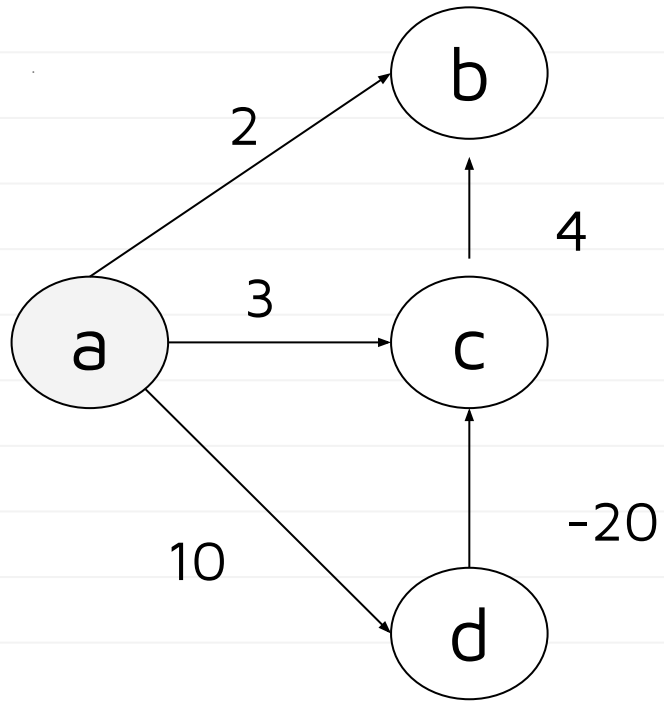
Example



Let's make another path that is **better** than 2.

Will the algorithm work for these weights? **Yes.**

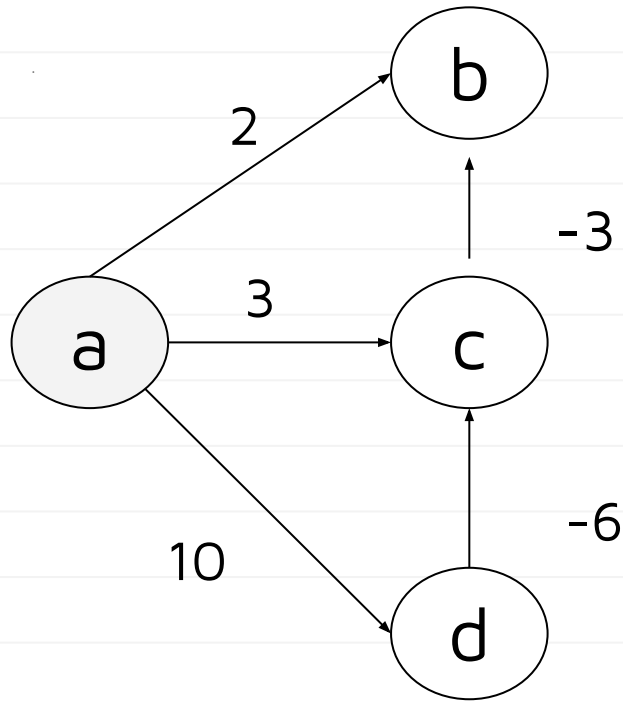
Example



Let's make another path that is **better** than 2.

Will the algorithm work for these weights?

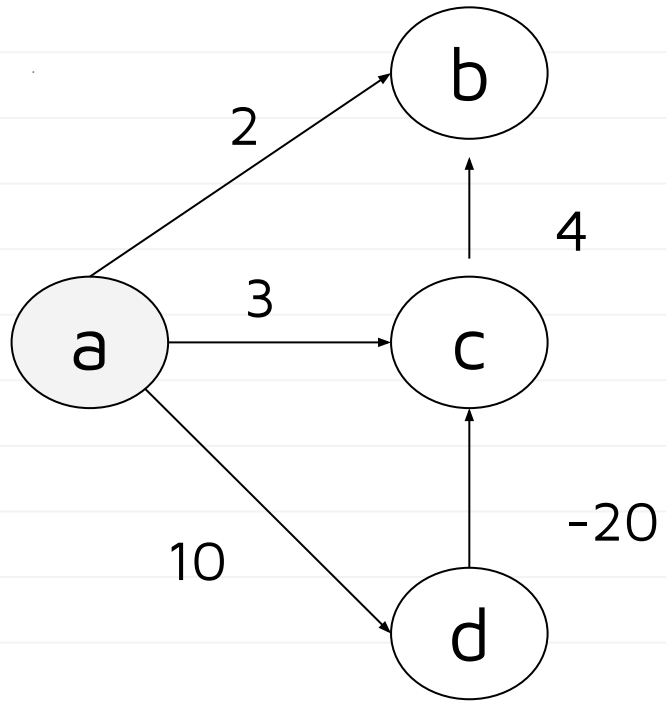
Example



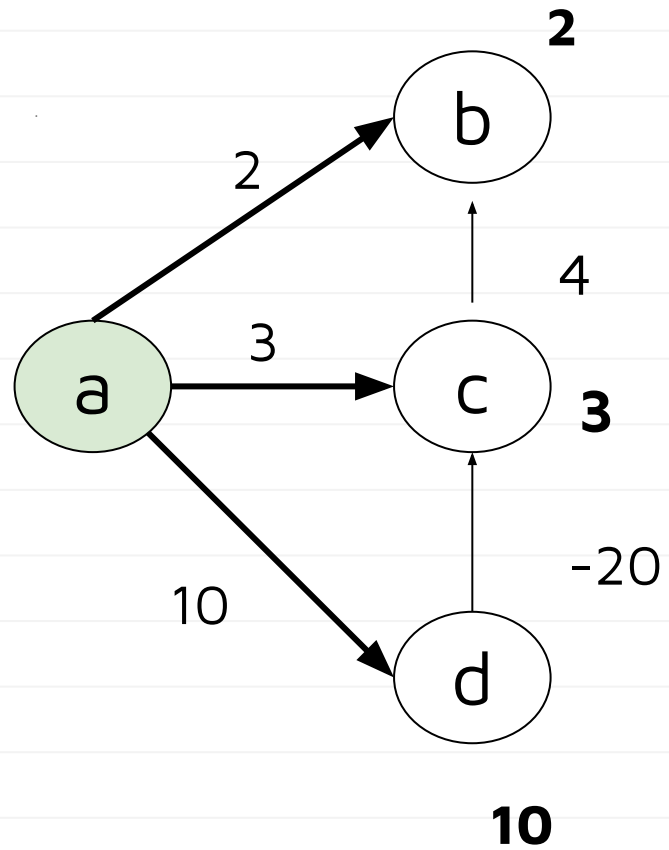
Let's make another path that is **better** than 2.

Will the algorithm work for these weights? **No.**

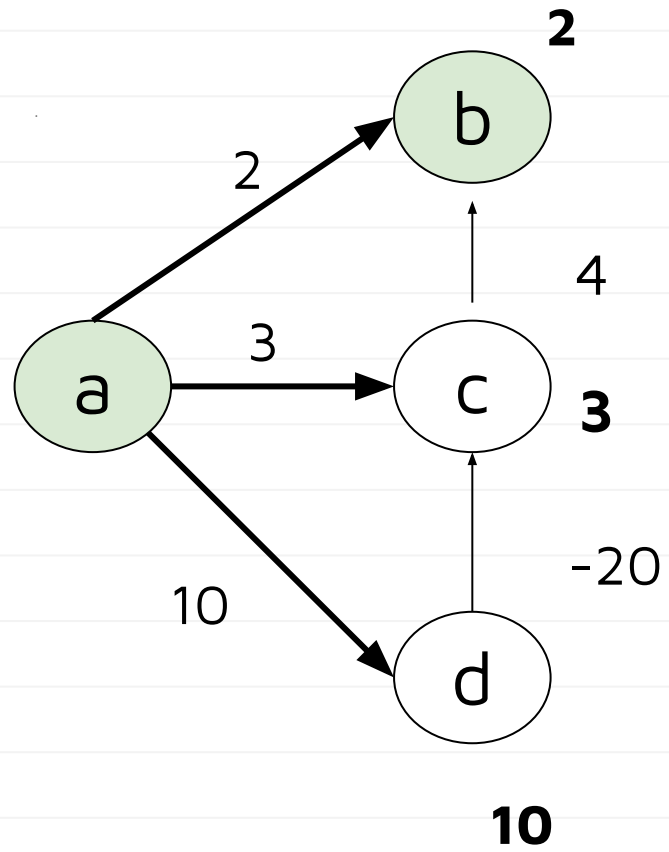
Example



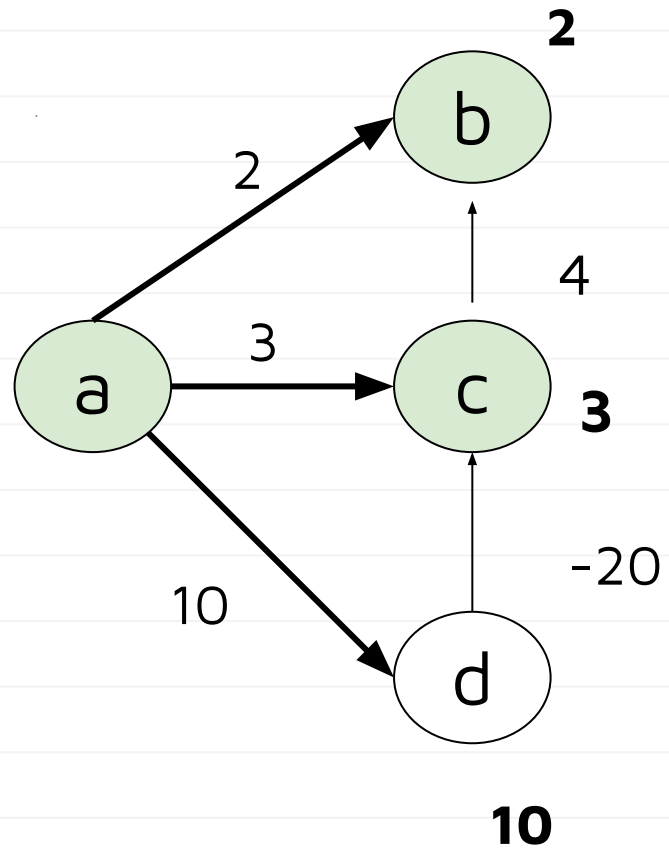
Example



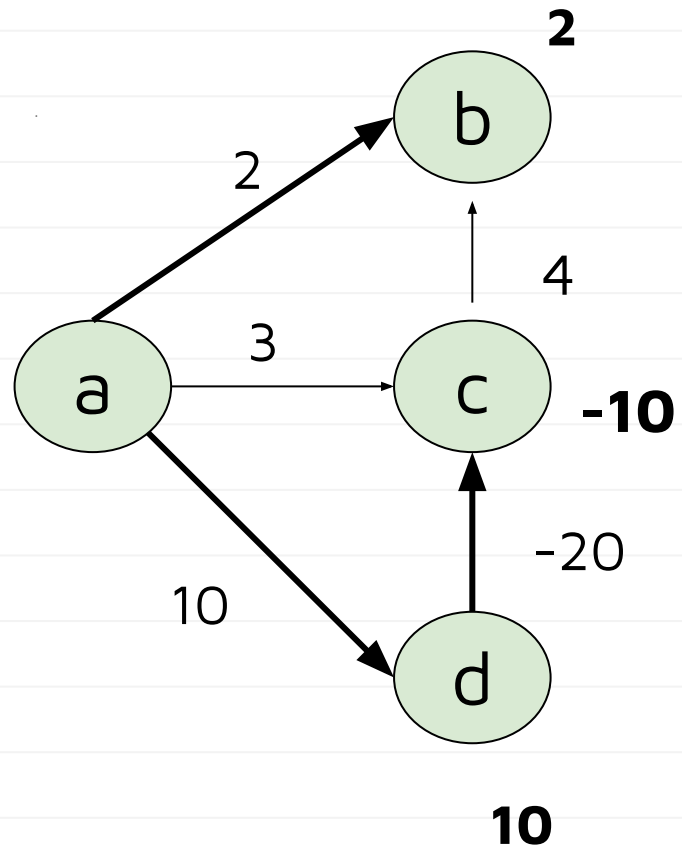
Example



Example



Example



Implementation

Outline of Dijkstra's Algorithm

```
def dijkstra(graph, weights, source):  
    est = {node: float('inf') for node in graph.nodes}  
    est[source] = 0  
    pred = {node: None for node in graph.nodes}  
  
    # empty set  
    C = set()  
  
    # while there are nodes still outside of C  
    #     find node u outside of C with smallest  
    #     estimated distance  
    C.add(u)  
  
    for v in graph.neighbors(u):  
        update(u, v, weights, est, pred)  
  
    return est, pred
```


Dijkstra's Algorithm: Naïve Implementation

```
1  def dijkstra(graph, weights, source):
2      est = {node: float('inf') for node in graph.nodes}
3      est[source] = 0
4      pred = {node: None for node in graph.nodes}
5
6      outside = set(graph.nodes)
7
8      while outside:
9          # find smallest with linear search
10         u = min(outside, key = est.get)
11         outside.remove(u)
12         for v in graph.neighbors(u):
13             update(u, v, weights, est, pred)
14
15     return est, pred
```

Dijkstra's Algorithm: Naïve Implementation

```
1  def dijkstra(graph, weights, source):
2      est = {node: float('inf') for node in graph.nodes}
3      est[source] = 0
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6      outside = set(graph.nodes)
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8      while outside:
9          # find smallest with linear search
10         u = min(outside, key = est.get)
11         outside.remove(u)
12         for v in graph.neighbors(u):
13             update(u, v, weights, est, pred)
14
15     return est, pred
```

inefficient

Priority Queue ADT

- Emergency Department waiting room operates as a priority queue
- Patients sorted according to seriousness, NOT how long they have waited



Priority Queues

- A **priority queue** allows us to store (key, value) pairs, efficiently return key with lowest value.
- Suppose we have a priority queue class:
 - `PriorityQueue(priorities)` will create a priority queue from a dictionary whose values are priorities.
 - The `.extract_min()` method removes and returns key with smallest value.
 - The `.change_priority(key, value)` method changes key's value.

Example

```
>>> pq = PriorityQueue({
    'w': 5,
    'x': 4,
    'y': 1,
    'z': 3
})
>>> pq.extract_min()
'y'
>>> pq.change_priority('w', 2)
>>> pq.extract_min()
?
```

Example

```
>>> pq = PriorityQueue({
    'w': 5,
    'x': 4,
    'y': 1,
    'z': 3
})
>>> pq.extract_min()
'y'
>>> pq.change_priority('w', 2)
>>> pq.extract_min()
'w'
```

Dijkstra's Algorithm: Priority Queue

```
def dijkstra(graph, weights, source):  
    est = {node: float('inf') for node in graph.nodes}  
    est[source] = 0  
    pred = {node: None for node in graph.nodes}  
  
    priority_queue = PriorityQueue(est)  
    while priority_queue:  
        u = priority_queue.extract_min()  
        for v in graph.neighbors(u):  
            changed = update(u, v, weights, est, pred)  
            if changed:  
                priority_queue.change_priority(v, est[v])  
  
    return est, pred
```

Heaps

- A *priority queue* can be implemented using a **heap**.
- If a **binary min-heap** is used:
 - `PriorityQueue(est)` takes $\Theta(V)$ time.
 - `.extract_min()` takes $O(\log V)$ time.
 - `.change_priority()` takes $O(\log V)$ time.

Time Complexity Using Min Heap

```
def dijkstra(graph, weights, source):  
    est = {node: float('inf') for node in graph.nodes}  
    est[source] = 0  
    pred = {node: None for node in graph.nodes}  
  
    priority_queue = PriorityQueue(est)  
    while priority_queue:  
        u = priority_queue.extract_min()  
        for v in graph.neighbors(u):  
            changed = update(u, v, weights, est, pred)  
            if changed:  
                priority_queue.change_priority(v, est[v])  
  
    return est, pred
```

Annotations:

- ← $\Theta(V)$ (points to the initialization of `est` and `pred`)
- ← (points to the `PriorityQueue` initialization)

Time Complexity Using Min Heap

```
def dijkstra(graph, weights, source):  
    est = {node: float('inf') for node in graph.nodes}  
    est[source] = 0  
    pred = {node: None for node in graph.nodes}  
  
    priority_queue = PriorityQueue(est)  
    while priority_queue:  
        u = priority_queue.extract_min()  
        for v in graph.neighbors(u):  
            changed = update(u, v, weights, est, pred)  
            if changed:  
                priority_queue.change_priority(v, est[v])  
  
    return est, pred
```

Annotations:

- $\Theta(V)$ for the initialization of `est` and `pred`.
- $\Theta(1)$ for the `update` function call.

Time Complexity Using Min Heap

```
def dijkstra(graph, weights, source):  
    est = {node: float('inf') for node in graph.nodes}  
    est[source] = 0  
    pred = {node: None for node in graph.nodes}  
  
    priority_queue = PriorityQueue(est)  
    while priority_queue:  
        u = priority_queue.extract_min()      O(log v )  
        for v in graph.neighbors(u):  
            changed = update(u, v, weights, est, pred)  Θ(1 )  
            if changed:  
                priority_queue.change_priority(v, est[v])  
  
    return est, pred
```

Time Complexity Using Min Heap

```
def dijkstra(graph, weights, source):  
    est = {node: float('inf') for node in graph.nodes}  
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            changed = update(u, v, weights, est, pred)  Θ(1 )  
            if changed:  
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    return est, pred
```

Time Complexity Using Min Heap

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    est = {node: float('inf') for node in graph.nodes}  
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    priority_queue = PriorityQueue(est)  
    while priority_queue:  
        u = priority_queue.extract_min()      O(log v )  
        for v in graph.neighbors(u):  
            changed = update(u, v, weights, est, pred)  Θ(1 )  
            if changed:  
                priority_queue.change_priority(v, est[v])  O(log v )  
  
    return est, pred
```

Time Complexity:

Time Complexity Using Min Heap

```
def dijkstra(graph, weights, source):  
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    priority_queue = PriorityQueue(est)  
    while priority_queue:  
        u = priority_queue.extract_min()      O(log v )  
        for v in graph.neighbors(u):  
            changed = update(u, v, weights, est, pred)  Θ(1 )  
            if changed:  
                priority_queue.change_priority(v, est[v])  O(log v )  
  
    return est, pred
```

Time Complexity: $O(V \log V + \quad)$

Time Complexity Using Min Heap

```
def dijkstra(graph, weights, source):  
    est = {node: float('inf') for node in graph.nodes}  
    est[source] = 0  
    pred = {node: None for node in graph.nodes}  
  
    priority_queue = PriorityQueue(est)  
    while priority_queue:  
        u = priority_queue.extract_min()      O(log v )  
        for v in graph.neighbors(u):  
            changed = update(u, v, weights, est, pred)  Θ(1 )  
            if changed:  
                priority_queue.change_priority(v, est[v])  O(log v )  
  
    return est, pred
```

Time Complexity: $O(V \log V + E \log V)$

Loop Invariant

- Assume that edge weights are **positive**.
- Before each iteration of Dijkstra's algorithm, both the *distance estimate* and the *predecessor* of the **first** node in the *priority queue* are **correct**.

Exercise

True or False: in Dijkstra's algorithm, a node's predecessor can be changed *after* it is *first* set.

A: True

B: False

C: Not sure yet

Exercise

True or False: in Dijkstra's algorithm, a node's predecessor can be changed *after* it is *popped* from the **priority queue**.

A: True

B: False

C: Not sure yet



Thank you!

Do you have any questions?

CampusWire!